Rule Book
RoboCup Humanoid Soccer League (HSL)

RoboCup Humanoid Soccer League Technical Committee
https://hsl.robocup.org/

(DRAFT 2026 Working Rules Document, as of 2026-06-11)

Questions or comments on these rules should be submitted via the following channels:

- https://github.com/RoboCup-HumanoidSoccerLeague/HSL-Rules/issues;
- Discord server: https://discord.gg/67upw6Nn, channel #rule-book;
- Email: hsl_tc@lists.robocup.org

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I Purpose and Scope of the RoboCup Humanoid Soccer League

This document defines the purpose, scope, and rules of the RoboCup Humanoid Soccer League for the international RoboCup competition, at the forefront of research, innovation, and education in humanoid soccer across all forms.

Vision and Mission Statement

The RoboCup Humanoid Soccer League (HSL) will drive innovative research that advances the software and hardware of autonomous robots, with a particular emphasis on robots deployed in real-time, dynamic, partially observable, and multi-agent environments. The HSL is well suited to advance *research*, *development*, and *education* in:

- Multi-robot systems (5+ robots) requiring decentralized coordination with limited communication over noisy channels.
- Robots that approach or approximate Human-like capabilities.
- A league that promotes research,
- Localization and state-estimation.
- Dynamic humanoid motor control.
- Real-time and on-board robot perception.
- Software engineering for autonomous robots.
- Hardware engineering for autonomous robots.
- Across topics, robot learning in all its forms.

In addition, the HSL aim to:

- Grow the community of humanoid soccer within RoboCup.
- Further education in robotics and is designed such that both teams with a primary focus in research and teams with a primary focus in education are able to participate.
- Encourage active sharing of software and hardware designs for league-wide collaboration.
- Measure the capabilities of the league against the 2050 vision of RoboCup.
- Drive the vision and direction of the rules to encourage good quality soccer between between evenly matched teams.

Core Vision and Requirements for Legal Standard Robot Platforms

The HSL encourages and welcomes the use of standard humanoid robot platforms available within the market to advance the state of humanoid robot soccer and the vision of the HSL.

With the HSL vision in mind, the core characteristics for standard humanoid platforms used within the HSL are:

1. Capable of dynamic motions such as fast walking, kicking a ball off the ground, and getting up from the ground;
2. Capable of running state-of-the-art AI neural network models for perception, decision-making, and control;
3. Sufficiently small and affordable to reasonably allow teams to fund multiple robots and travel with them to competitions;
4. Able to be programmed at a low level of control;
5. Well-documented to enable effective use and development by teams.

Core Vision and Requirements for Constructed Robots

The HSL equally encourages and welcomes the use of fully custom built or modified humanoid robot platforms. To create a welcoming and fair environment for all robot platforms, the HSL ensures the following:

1. Both store-bought and custom-built robots can participate in a fair competition without risking damage to their robots.
2. The tournament is designed such that games are interesting for all participating teams and match-ups are fair.
3. The tournament is designed such that all currently existing teams are able to participate.
4. Details about hardware and software of the robots is made available to teams and organizers to ensure a fair competition and encourage scientific exchange.
5. League resources are distributed such that both store-bought and custom-built robots equally benefit from them.
6. Robots are designed with the goal of RoboCup in mind, thus restricting the allowed sensors where possible to humanoid sensors. Exceptions to this rule can be made if it benefits scientific progress.

II Notes and modifications

Notes on the Laws of the Game

Official Languages

RoboCup Humanoid Soccer League Technical Committee publishes the Laws of the Game in English.

Measurements

If there is any divergence between metric and imperial units, the metric units are authoritative.

Gender-Neutral Language

References to human referees, assistant referees, and officials are formulated using gender-neutral language. References to robot players use the neutral pronoun *it* and *its*.

General Modifications

Subject to the agreement of the member association concerned and provided the principles of these Laws are maintained, the Laws may be modified in their application for regional matches. Any or all of the following modifications are permissible:

- size of the field of play
- size, weight and material of the ball
- width between the goalposts and height of the crossbar from the ground
- duration of the periods of play
- substitutions

III Laws of the Game

§ 1 The Field of Play

The competition consists of three distinct divisions, categorized specifically by the maximum physical height and weight of the robotic platforms.

To ensure a fair and proportionate competitive environment, the field dimensions, number of players and ball size will vary to align with the specific requirements of each division (see § 3 for more details).

§ 1.1 Field surface

The matches are played on artificial turf with a height between approximately 20 to 30 mm, except for the small division, which has a height between 8 and 12 mm, also approximately.

The color of artificial turf must be green. No particular shade is required, but the green must contrast well with the field markings and the ball and should not be very dark.

§ 1.2 Field markings

The field of play must be rectangular and marked with continuous lines. These lines belong to the areas of which they are boundaries.

The color of the field markings should be white, whether applied by tape, paint, or made from white turf.

The two longer boundary lines are called touchlines (or sidelines). The two shorter lines are called goal lines. The field of play is divided into two halves by a halfway line, which joins the midpoints of the two touchlines.

A team’s half is the half of the field that is closest to the team’s own goal, where it primarily defends against opposing attacks.

The center mark is at the midpoint of the halfway line. A circle is marked around it, defining the center circle, a region used for kick-offs and other specific plays.

The goal area and the penalty area are defined using lines drawn at right angles to the goal line.

All lines must be of the same width, which must be between 5 and 12 cm.

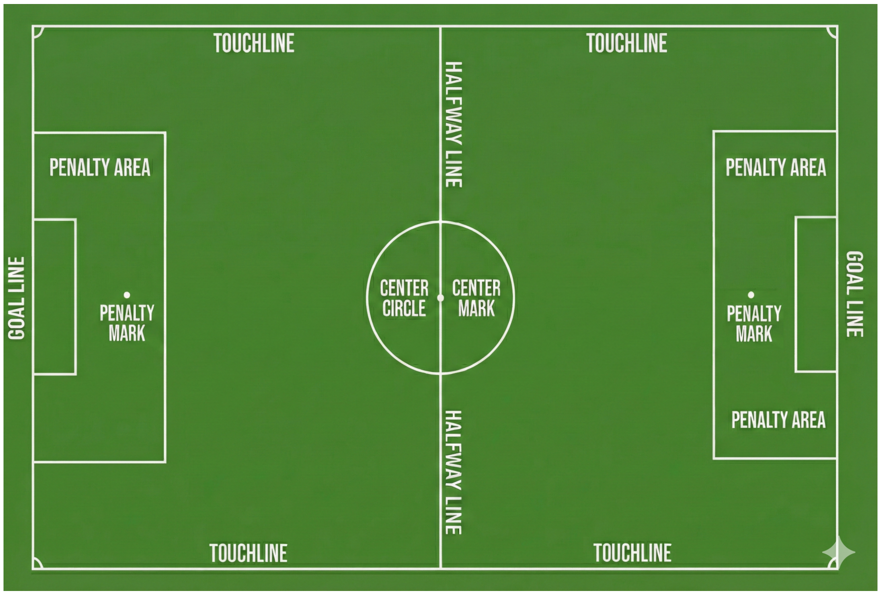


Figure 1: Annotated diagram illustrating the standard markings and key areas of a soccer field.

Figure 1 shows the field markings with the lines and its names.

§ 1.3 The goal area

Two lines are drawn at right angles to the goal line; the length and distance from the midpoint of the goal line depend on the division.

These lines extend into the field of play and are joined by a line drawn parallel with the goal line. The area bounded by these lines and the goal line is the goal area.

§ 1.4 The penalty area

Two lines are drawn at right angles to the goal line; the length and distance from the midpoint of the goal line depend on the division.

These lines extend into the field of play and are joined by a line drawn parallel with the goal line. The area bounded by these lines and the goal line is the penalty area.

Field	Width (meters)	Length (meters)
S-Field (small field)	6	9
M-Field (middle field)	6 to 9	9 to 14
L-Field (large field)	9 to 14	14 to 22

Table 1: Approximate field sizes.

Within each penalty area, a penalty mark is made.

§ 1.5 The corner arc

A quarter circle from each corner is drawn inside the field of play. The radius of the arc is 0.5m for the M-Field and 1.0m for the L-Field. The S-Field does not have corner arcs.

§ 1.6 Dimensions

There are three field sizes that can be used for the proposed divisions, described in Table 1.

Table 2 shows the dimensions of the S-, M- and L-Fields. Diagrams of the three soccer fields are shown in Figures 2 to 4. Note that the Middle Division can be played on the S-Field or the M-Field, while the Large Division can be played on both the M-Field and the L-Field. More detailed technical drawings are provided in Section A.

Note that measurements are from the outside of the lines as the lines are part of the area they enclose (as it is in the FIFA Rules).

§ 1.7 Goals

A goal must be placed on the center of each goal line. A goal consists of two vertical posts equidistant from the corners and joined at the top by a horizontal crossbar.

The goalposts and crossbar must be made of wood, metal, or other approved material and must not be dangerous to the players.

The goalposts and crossbar of both goals must be of the same shape, which must be square, rectangular, round, elliptical, or a hybrid of these options.

need a figure how goalposts must be placed on the goal line

The goalposts and crossbar must have the same width, which is not less than 7 cm and does not exceed 12 cm. The goalposts and the crossbar must be white. Goal nets and any net supports may be white, gray, or black.

Goals must be anchored securely to the ground.

ID	Description	S-Field	M-Field	L-Field
	(all values in meters)	*(former KidSize and SPL)*	*(former AdultSize)*	*(similar to MSL)*
A	Field length	9.0	14.0	22.0
B	Field width	6.0	9.0	14.0
C	Goal length (depth)	0.5 to 1.0	0.7 to 1.2	1.0 to 2.0
D	Goal width	1.8 to 1.9	2.4 to 2.6	2.6 to 3.1
E	Goal Area length	1.0	1.0	1.0
F	Goal Area width	3.0	4.0	5.0
G	Penalty Area length	2.0	3.0	3.5
H	Penalty Area width	4.0	6.0	7.0
I	Penalty Mark distance	1.5	2.0	2.5
J	Center Circle diameter	1.5	3.0	4.0
K	Border strip width (min.)	1.0		
L	Corner Arc radius	none	0.5	1.0
	Line width	0.05		0.12
	Penalty and center mark size	0.10		0.15

Table 2: Exemplary field dimensions by field type, in meters.

Off-the-shelf goals may be used if they satisfy these requirements.

§ 1.8 Technical Area

The *Technical Area* is the designated zone located on the sidelines. The GameController operator is positioned centrally, aligned with the mid-field line. At a minimum, this area includes the official tables and the space behind them. The boundaries may be extended at the committee's discretion through visual markings or official announcements during the event.

Each team is assigned a dedicated space on either side of the Game Controller operator:

- **Home Team:** Occupies the space to the **right** of the operator.
- **Away Team:** Occupies the space to the **left** of the operator.

This space is intended for all activities necessary for playing, including (but not limited to) operating laptops, storing spair robots, and performing robot repairs.

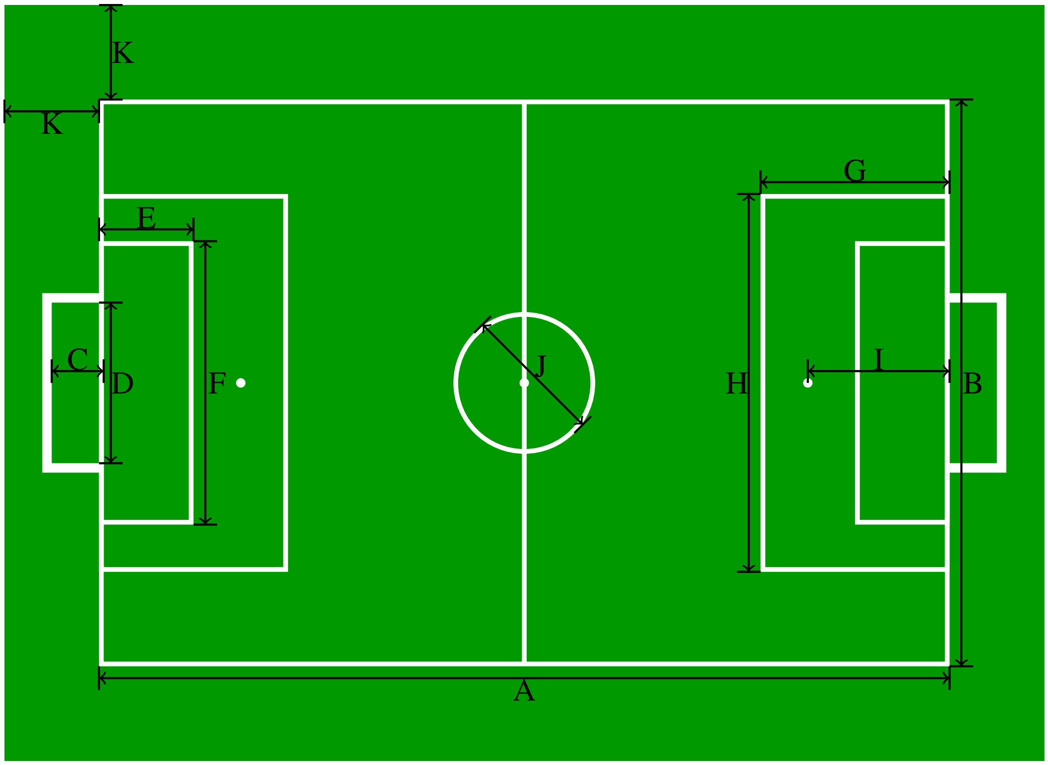


Figure 2: Schematic diagram of the small soccer field – S-Field (scale: 1/80)

Division	Width (meters)	Height (meters)	Depth (meters)
Small	1.8 to 1.9	1.2 to 1.3	0.5 to 1.0
Middle	2.4 to 2.6	1.5 to 1.9	0.7 to 1.2
Large	2.6 to 3.1	1.8 to 2.0	1.0 to 2.0

Table 3: Allowed ranges for goal dimensions.

§ 1.9 Definition of Inside and Outside

A robot is considered *inside* a region of the field if any part of its body is touching the ground inside the region or its boundary lines. It is considered *outside* the region only when no part of its body is touching the ground inside the region and its boundary lines.

The ball is considered *inside* a region of the field if any part of it overlaps or touches the boundary lines that define that region, or if it is fully contained within the region, in the air or on the ground. It is considered *outside* the region only when no part of it, or its downwards projection, remains within or on the boundary lines of that region. This definition applies to any designated area of the field, except the goal.

Exception for the goal: The ball is only considered *inside* the goal (and thus a goal is scored) if, and only if, the entire ball has wholly crossed the goal-side edge of the goal line *between the goal*

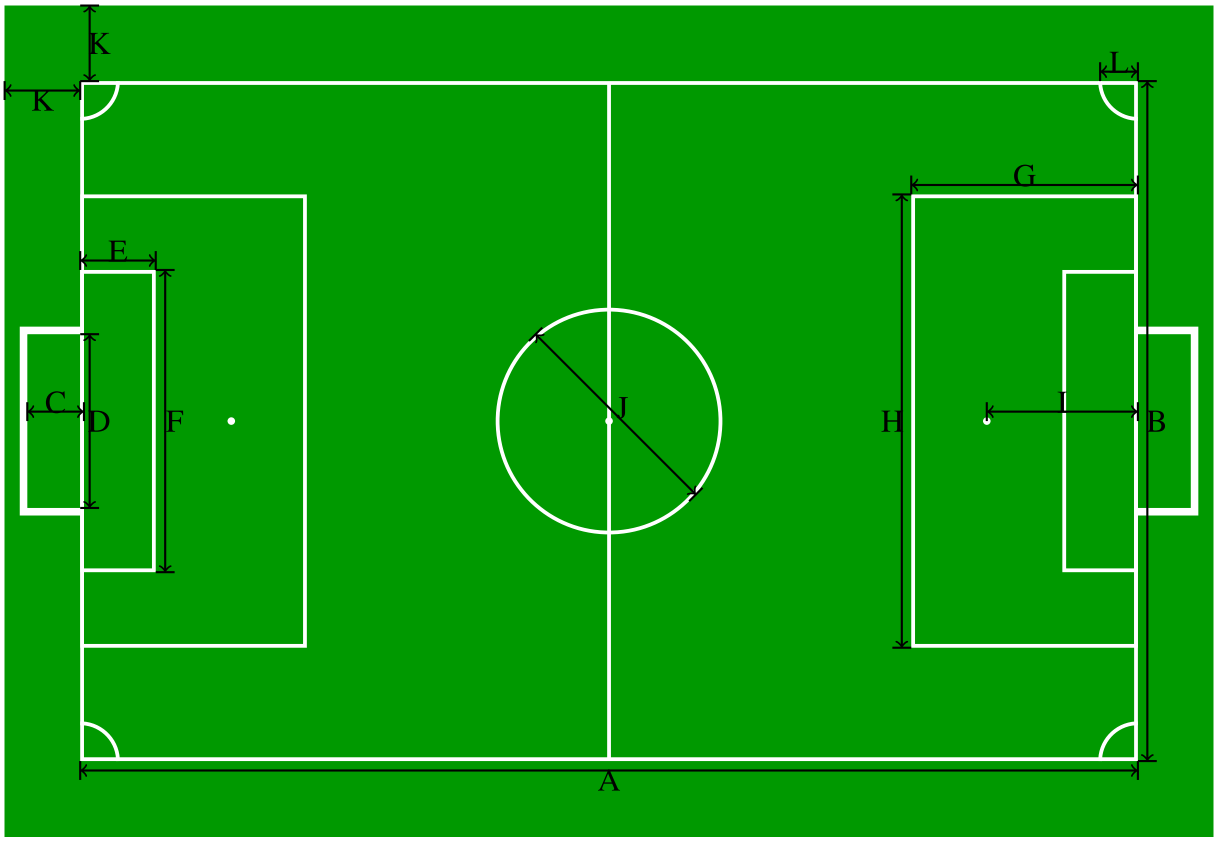


Figure 3: Schematic diagram of the medium soccer field – M-Field (scale: 1/100)

posts and below the crossbar, i. e. when no part of the ball remains on or above the goal line, and the entire ball is within the volume enclosed by the goal frame and the net.

§ 1.10 Venue Setup

Fields may be located close to one another. Barriers will not necessarily be constructed between adjacent fields to block the robots from seeing other fields, goals, or balls. However, barriers will be constructed to block sight between any fields that are not located at least three meters apart. Hence, for each side of a field that is adjacent to another field, either barriers will separate the fields or at least $3\,\mathrm{m}$ will be between the carpet of adjacent fields.

§ 1.11 Lighting Conditions

The HSL does not mandate specific or controlled lighting conditions for a match venue, but adherence to a few guidelines is required.

It is expected that the venue provides reasonable lighting suitable for general visibility, approximating natural daylight (e. g. indoor with artificial lighting, outdoor with natural lighting, or a combination of both).[1]

[1] The absolute minimum requirement is that most of the field is at least $300\,\mathrm{lx}$ (preferably $400\,\mathrm{lx}$).

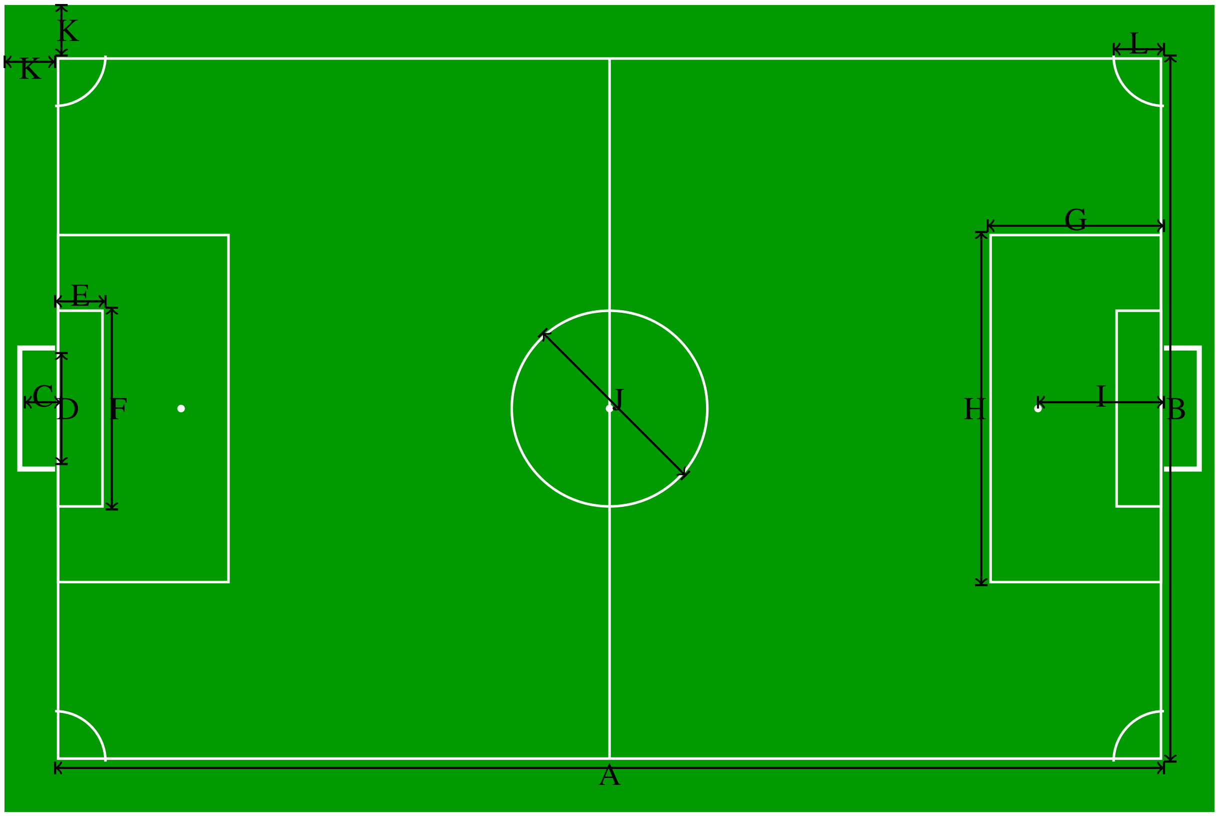


Figure 4: Schematic diagram of the large soccer field – L-Field (scale: 1/150)

The light must be predominantly white. Colored lighting that significantly changes the perceived color of the field or ball must be avoided.

Apart from these requirements, the specific lighting conditions depend on the actual venue. It may include variations such as glare, brightness, shadows, or mixed lighting conditions that can change throughout the match. Fields should be placed near or under windows where possible.

Teams participating in the HSL are encouraged to design their robots to handle a variety of typical lighting environments that may be encountered during a match. Natural and non-natural light must be free to reach the field. The technical committee can delimit a zone near the field where humans must not stand and where any items blocking the light sources are forbidden.

§ 2 The Ball

§ 2.1 Qualities and measurements

All balls must be spherical, made of or resembles the weight, form, movement characteristics and appearance of leather or other suitable material.

The balls for each division[2] are defined in Table 4.

Division	Ball Type
Small	FIFA Mini Ball
Middle	FIFA size 3 or 4
Large	FIFA size 5

Table 4: Balls used in each division

§ 2.2 Replacement of a defective ball

If the ball bursts or becomes defective during the course of a match: the match is stopped. It is restarted according to the drop ball rule (see § 8).

If the ball bursts or becomes defective whilst not in play at a kick-off, goal kick, corner kick, free kick, penalty kick or throw-in: the match is restarted accordingly.

If the ball becomes defective during a penalty kick or penalties (penalty shoot-out) as it moves forward and before it touches a player, crossbar or goalposts, the penalty kick is retaken.

The ball may not be changed during the match without the referee's permission.

§ 2.3 Additional balls

Additional balls which meet the requirements of § 2 may be placed around the field of play and their use is under the referee's control.

[2]The TC and OC will decide on only one type of the ball per division that will be used by all teams during the 2026 competition.

§ 3 Teams and Players

§ 3.1 Divisions and Configurations

A team belongs to one division, which is one of Small, Middle and Large. Main competition matches only take place between members in the same division.

Teams in each division can choose to play in two team configurations: Foundation - with a smaller number of players; and Advanced - with a larger number of players. If in a game one of the teams chooses to play in a Foundation configuration, then both teams must play in this configuration to ensure fairness. The maximum number of players in a game for each configuration is shown in Table 5.

In the Middle division, starting from the quarter-finals onwards, all matches will be played in the Advanced configuration.

The division and the configuration preference of a team must be submitted before the competition. The division must not change during the competition itself. The division must be submitted at the time of team registration. In addition, each team shall submit its preferred configuration prior to the competition, by a deadline specified by the organizers, in order to enable proper field planning. A team may request to play in a different configuration during the competition by submitting a request to the organizers. Approval of such a request is at the sole discretion of the organizers.

§ 3.2 Number of Players

A match is played by two teams, each with a **maximum** number of players determined by the division the game takes place in and the configuration level of the playing teams, as illustrated by the following table:

Division	**Foundation teams**	**Advanced teams**
Small	4	7
Mid	3	5
Large	3	5

Table 5: Number of players

At most one player per team on the field may be designated as *goalkeeper*, the others are all *field players*. When playing at full strength, a team must have a *goalkeeper* on the field.

§ 3.3 Number of Substitutes

In addition, each team may prepare *substitute players* outside the field. The number of substitute players is not limited. A *substitute player* may be substituted in to become a *field player* or *goalkeeper*.

§ 3.4 Substitution Procedure

To replace a player with a substitute, the following conditions must be observed:

- The substitution may only occur during the `initial`, `set` and `finished` states (after any penalties are assigned). Robots subject to a standard removal penalty may also be substituted during a brief stop. See § 7 for details on the game states.
- The team requests the substitution to the head referee. The substitiution can only be made if the head referee approves.
- If the player being substituted is on the field, it must first be removed from the field.
- The substitution is input in the GameController.
- If the player being substituted is not subject to a standard removal penalty, the substitute can either enter the game immediately or remain in the team area for further maintenance and enter at any later time. In either case, when the substitute is ready to enter, it must be given to a robot handler for placement on the standard re-entry point (see § 12), and the GameController operator must be informed. The robot will join the game immediately after.
- If the player being substituted is subject to a standard removal penalty, the substitute inherits its penalty state and penalty timer. It then behaves like any other penalized robot, and re-enters the game according to the corresponding rules (see § 12).
- Robots that have been sent off cannot be replaced by a substitute (see § 12).

§ 3.4.1 Changing the Goalkeeper Any of the other players may change places with the goalkeeper, provided that:

- The referee is informed before the change is made.
- The change is approved by the head referee.
- The change is requested during the `initial`, `set` or `finished` state.

§ 3.5 Robot Players

§ 3.5.1 The Design of the Robots Robots participating in the HSL must have a human-like body shape with a torso, head, two arms, and two legs, as well as human-like symmetry and proportions regarding sizes of the body parts and weight distribution.

The robots must be able to stand upright on their feet, to walk on their legs and to be able to recover from a fall (get back to a standing position).

The only allowed modes of locomotion are bipedal walking, running, and jumping. Soccer related movements such as dribbling, kicking, or other forms of ball handling are also allowed.

The design of the robot's arms, including their length and placement, shall permit arm use and behaviors that are reasonably comparable to those of humans. Examples of permitted uses include assisting in getting up after a fall or picking up and throwing the ball (where otherwise allowed by the rules).

Arm configurations that enable behaviors significantly different from those of humans are not permitted. In particular, robots must not use their arms to provide continuous support for locomotion, such as walking on arms or using arms as additional legs.

§ 3.5.2 Size Restrictions All robots participating in the HSL must comply with the following restrictions:

The length of the legs H_{leg}, including the feet, satisfies $0.35 \cdot H_{top} \leq H_{leg} \leq 0.7 \cdot H_{top}$, where H_{top} is the height of the top of the robot. The length of the leg is measured from the first rotating joint where its axis lies in the plane parallel to the standing ground to the tip of the foot.

A classic piece of human anatomy and art history, Leonardo da Vinci's "Vitruvian Man" famously depicts a man whose arm span is equal to his height, creating a 1:1 ratio. Therefore, the arm span, A_{span}, including the hands, should satisfy $0.8 \cdot H_{top} \leq A_{span} \leq 1.2 \cdot H_{top}$.

Based on H_{top}, the following size restrictions apply for each division:

- Small: $H_{top} \leq 1.1$ meters;
- Middle: $H_{top} \leq 1.25$ meters;
- Large: $H_{top} \leq 1.9$ meters.

H_{top} is defined as the height of the robot when standing upright (with fully extended knees). H_{top} is measured with the head of the robot oriented in such a way that it is tilted to either its maximum upwards tilt angle or the horizon line, whichever is lower.

The height of the head H_{head}, including the neck, satisfies $0.1 \cdot H_{top} \leq H_{head} \leq 0.3 \cdot H_{top}$. H_{head} is defined as the vertical distance from the axis of the first arm joint at the shoulder to the top of the head.

§ 3.5.3 Weight Restrictions The robot's Body-Mass Index (BMI) is defined as follows: $\mathrm{BMI} = \frac{M}{{H_{top}}^2}$, where M is the mass of the robot in kg and H_{top} its height in meters.

The Body Mass Index (BMI) of the robot should be: $5 \leq \mathrm{BMI} \leq 30$.

The following weight restrictions apply for each division:

- Small: Weight$\leq$ 15 kilograms;
- Middle: Weight $\leq$ 25 kilograms;
- Large: Weight $\leq$ 80 kilograms.

§ 3.6 Approved Standard Platforms

The following commercially available robotic platforms are approved for participation in the HSL.

The following are the list of pre-approved standard platforms that require no modifications:

Manufacturer	Model	Restrictions
Aldebaran	Nao V5, and Nao V6	
Robotis	DARwIn-OP	
Booster	T1, and K1	
Fourier	GR1, GR1-Pro, and GR2	
Unitiree	G1++, H1++	

Additional humanoid robot platforms can be approved upon request. Please send your request for approval to `hsl_tc@lists.robocup.org`.

§ 3.7 Hardware

Modifications or additions to the robot hardware are permitted as long as the resulting robot does not become non-compliant with these rules. No off-board sensing or processing systems are permitted.

§ 3.8 Sensors

Teams participating in the HSL competitions are encouraged to equip their robots with sensors that have an equivalent in human senses. These sensors must be placed at a position roughly equivalent to the location of the human's biological sensors.

The sensors and their placement shall be chosen such that they allow the robot a spatial perception similar to humans. The sensors are evaluated along the following two general guidelines:

1. *Foster and encourage research and development towards human-like perception capabilities.* Sensors aiming to directly emulate human senses, like a camera or a microphone. Use of such sensors is explicitly *encouraged*.

2. *Enable research and development under the constraints of the current state of the art in technology and research, as long as this does not undermine current research efforts as declared in point 1.*
Sensors that compensate for current shortcomings in technology and state of research, like one-dimensional distance sensors with limited range, or two vertically arranged cameras in the robot NAO.

Generally, *passive sensors* are preferred to *active* sensors that actively emit signals.

§ 3.8.1 Intrinsic Perception and Proprioception Any sensing capability aiming at measuring the internal state of the robot is permitted. This includes temperature, current consumed by the motors, joint positions, etc.

§ 3.8.2 Visual Perception The visual sensors, e.g., cameras of the robot shall be arranged such that the combined visual field is *contiguous* and limited to dimensions similar to a human, which corresponds to limitations in the opening angle: horizontal $\leq 220°$ and vertical $\leq 160°$. Please note that these ranges may be adjusted to more human-like viewing angles in the future.

The visual sensors shall be located in the head of the robot.

The combined dynamic visual field that can be observed by the robot solely by moving its cameras (similar to human eye movements) and the head is limited to horizontal $\leq 340°$ and vertical $\leq 220°$.

The cameras are restricted to visual information in the range of the light visible to humans.

The cameras can provide dense visual information, such as a rasterized image, or sparse information, such as visual events (event cameras).

§ 3.8.3 Visual (dense) Depth Perception Passive integrated devices that provide dense depth information, such as stereo cameras, are permitted and encouraged.

Active integrated devices that provide depth information, such as cameras with an active infrared projector or time-of-flight cameras, are permitted but discouraged. Their use might be prohibited in the future.

§ 3.8.4 Orientation Sensors providing information regarding the robot's orientation in space with relation to the ground are permitted. This includes sensors such as gyrometer, accelerometer, as well as integrated inertial measurement units (IMU), as long as they provide only relative measurements and no measurements of the absolute direction, such as a compass.

An IMU with an integrated compass can be used as long as the compass not used.

The sensors can be placed in the head and/or in the torso of the robot.

§ 3.8.5 Sound Perception The sound sensors, e.g., microphones, shall be placed in the head of the robot. The number of microphones is not limited.

§ 3.8.6 Haptic Sensing Any passive sensor allowing haptic measurement is permitted, such as force sensors, touch sensors, buttons/bumpers, capacitative touch sensors.

Haptic sensors can be placed at any location of the robot's body and are not limited in number.

§ 3.8.7 Distance Sensing Active and passive sensors for one-dimensional distance sensing with a limited range are permitted if their use is limited to compensate for shortcomings in the spatial awareness of the robot in the close proximity.

The number of such distance sensors is limited to 4.

The use of distance sensors is discouraged and might be prohibited in the future.

§ 3.8.8 Summary The Table 6 summarizes the approved sensors. The use of listed sensors that are considered human-like is encouraged. The use of listed sensors that are not considered human-like is accepted, but discouraged and might be prohibited or limited in the future.

§ 3.9 Communication

Robots participating in the HSL competitions must act autonomously during a game.

Any use external power supply, teleoperation, remote control, remote processing, remote sensing, or remote brain of any kind is prohibited.

Communication is only allowed among robots on the field, between the robots and the referees, and between the robots and the GameController.

§ 3.9.1 Acoustic and Visual Communication The robots are permitted to communicate through visual signaling (e. g., gestures) and / or acoustic signaling using the robot's built-in microphones and speakers.

Sensor Type	Human-Like	Comments
RGB camera	yes	opening angle limit: horizontal $\leq 180°$, vertical $\leq 140°$
Stereo camera	yes	
Event camera	yes	
Active RGB-D based on infrared projection	no	discouraged
Active RGB-D based on time of flight (TOF)	no	discouraged
Microphone array	yes	
Gyro	yes	
Accelerometer	yes	
Compass	no	magnetic measurement sensors are not allowed
Integrated inertial measurement units (IMU)	yes	no compass or must be disabled and not used
Force sensors	yes	
Touch sensors	yes	
Buttons / Bumpers	yes	
Capacitive touch sensors	yes	
Near range infrared sensor	no	
Sonar / ultrasound sensors	no	
1D LIDAR and laser range sensor	no	with limited range
Intrinsic sensors	yes	temperature, current, joint positions, etc.

Table 6: Summary of approved sensor types.

Communication that causes excessive discomfort to an audience, affects the safety of an audience, or violates normal playing rules is not permitted.

§ 3.9.2 Wireless Communications: The robots are permitted to communicate through WiFi using builtin wireless adapters.

A field is equipped with a wireless access point that must be used for any wireless communication during the game. Only the teams involved in the game are permitted to use the access point during a match.

Each team will be assigned a range of IP addresses that can be used both for their robots and their computers. The network configuration (e. g. IP addresses, channels, SSIDs, and required encryption) of the fields will be announced at the competition site.

Wireless equipment and access points will be provided by the organizers of the competition. No other $2.4\,\mathrm{GHz}$ or $5\,\mathrm{GHz}$ radio equipment (including cellular phones and/or Bluetooth devices) is allowed to be used close to the field.

§ 3.9.3 Wireless Robot-To-Robot Communication: Wireless robot-to-robot communication between the robot players is allowed and must use the access points provided by the event organizers. Direct wireless communication between the robots (using the so-called ad-hoc mode) is prohibited.

Messages must be sent via UDP broadcast. Each message's payload size must not exceed $512\,\mathrm{B}$.

Each team will be allocated a single UDP port on which to send/receive their team messages. Specifically, a team's port will be 10000 plus that team's GameController number.

Unicast communication between robots is prohibited.

Each team is allowed to send in total a maximum of 12000 messages during a complete game. This limit can change during a game only in these cases:

- For each minute of additional time added by the *Allowance for time lost* rule (see Section 7), the limit is immediately increased by 600 packets.
- If the game proceeds to extra time (see Section 10), the limit is increased by 6000 packets before the beginning of the first half of extra time.

Only messages sent by the robot players in the following game states count towards the communication limit: `ready`, `set` and `playing`. Any messages sent during other game states do not count towards the teams message budget. Limits do not apply during penalty kick shoot-out.

The number and the size of the messages sent by each team is monitored by the GameController.

A violation of the permitted maximal size of a message or the maximal number of messages is subject to penalty (see Section 12).

The limit of allowed packets will be lowered in future competitions to encourage a progress towards human-like communication and coordination.

§ 3.9.4 GameController Communication: The GameController will communicate at 2 packets per second.

Robots are required to respond to GameController messages and send status update packets to the GameController.

The format for the GameController messages and for the responses is defined in the official GameController[3]

§ 3.9.5 Debug Communication: Team specific debug information may be sent to an external computer owned by the team. Each robot may send debug information in a single UDP packet to a single team device on the wired network. The robot may sent at most 1 packets per second. The size of a single packet is not explicitly restricted, but must be within the maximum single UDP packet size (typically 64kB).

§ 3.10 Safety

§ 3.10.1 Dangerous Equipment A player must not use equipment or wear anything that is dangerous to himself or another player.

§ 3.10.2 Physical Stop Robot handlers must be able to immediately and physically bring a robot to a safe state at any point in the game (see § 5).

It is the responsibility of the teams to ensure that a robot can be physically stopped by their handler in case of emergency.

The team must be able to demonstrate their ability to physically stop their robots on request at any time during the competition.

In order to ensure that a robot can be physically stopped a robot at any time, the robots can be equipped with dedicated safety features.

Passive Features The robot can be equipped with passive components that allow for it to be physically restrained without danger to the human handler. Examples of such components are grips or handles on the back of the robot, and design features of the robot's body that allows a human to safely hold the robot.

[3] `https://github.com/RoboCup-HumanoidSoccerLeague/GameController/blob/master/game_controller_msgs/headers/RoboCupGameControlData.h`

Active Features The robot can be equipped with active components that makes the robot to immediately stop all movements and assume a safe posture. Examples for such components are a physical button on the robot, emergency switch on the robot, remote emergency button.

Remote Emergency Button Robots can be equipped with a remote emergency button. A remote emergency button is a device that is not physically attached to the robot and can send a remote signal to the robot causing it to stop all movement and assume a safe posture. Each emergency stop button can only be connected to one specific robot. The remote emergency button cannot perform any additional functions.

§ 3.10.3 GameController Stop The robots must comply with the GameController protocol and immediately assume a safe pose when the command is sent by the GameController as formulated in § 3.9.2 and § 7.1.7.

§ 4 The Players' Equipment

§ 4.1 Team Markers

Robot bodies must be colored mostly with a neutral color, such as black, gray, white, or silver, and be non-reflective.

Robots must use colored jersey shirts as team markers. A jersey is a tank top-style shirt that may start at the neck and go down to the waistline, and does not cover the arms of the robots. It must cover at least 50% of the upper body of the robot.

Each robot has a unique jersey number from the set $\{1, 2, 3, \ldots, 20\}$.

Jerseys must have a primary color that comprises more than half of the jersey. The primary color must be recognizable from the front and back.

Jerseys should not contain distracting patterns that could be confused with other elements of the soccer field, such as field lines or the ball.

Jersey material must be non-reflecting, non-shiny, and non-glittery. Jerseys may be manufactured from fabric and fine mesh.

The two teams must wear colors that distinguish them from each other and also the referee and the assistant referees.

Each team must provide two jersey designs in distinctly different solid colors. The jersey color must clearly contrast with any uncovered parts of the robot to ensure reliable identification.

Robots must display clear identification numbers (player number): a large number on the back and a smaller number on the front.

The goalkeeper robot must be uniquely marked so that it can be easily distinguished from the other robots on the same team by the referees. Some possible examples of markings:

- A distinct jersey color. The goalkeeper is allowed to wear a jersey with a different primary color from the rest of the team, either by wearing the team's second kit or by introducing a new color entirely. If the goalkeeper does use a distinct jersey color, it must be different from the field players of both teams, and it should also be different from the opposing goalkeeper if possible. Jersey numbers must still be unique among all the robots of one team currently on the field.
- The jersey number. Informing the referees before the game starts of which jersey number will act as the goalkeeper is considered a sufficient marking. The goalkeeper may use any of the allowed jersey numbers.

Sponsorship logos. Sponsorship logos on the jerseys are allowed on the arms and legs of the robots, as well as the jerseys. Logos on the arms and legs must not cover more than 25% of the limb, and logos on jerseys must not occupy more than 50% of the surface, ensuring that a suitable portion of the robot remains in the respective primary color.

Logos, like the other aesthetic elements, must not interfere with the player numbers, and they must not include distracting patterns and colors that could be confused with other elements of the soccer field, such as the field lines or the ball.

Approval. Jersey designs must be submitted for approval prior to the competition by a deadline specified by the organizers. Jerseys shall be reviewed and approved by the technical committee prior to the competition.

§ 5 Human Officials

Each match is run by a number of human officials. They are distinguished in two categories: referees, neutral persons who have the the duty to enforce the rules of play and ensure the game proceeds smoothly (including the GameController operator); and team officials, who belong to either of the teams and perform specific tasks for that team.

The referees are assigned before the game: example modalities are by scheduling in official games or by common agreement in friendly games.

As for team officials, each team selects theirs officials before the start of the game and communicates their selection to the head referee during the pre-game meeting. One person may cover multiple roles, unless otherwise specified.

§ 5.1 Equipment of human on the field

All humans on the field must not wear white or green footwear, socks, or any other foot covering in these reserved colors.

All clothing worn on the field, including referee jerseys, must not contain distracting patterns or designs that could be confused with elements of the soccer field, such as field lines or the ball.

They may also have equipment to communicate with each other, such as a headset or flags, if available.

The referee jersey provided by the event is recommended for all referees but is not mandatory. The head referee must also be equipped with a whistle, a coin, and yellow and red cards.

§ 5.2 Head Referee

The head referee is the principal authority of the game and takes all decisions regarding the flow of the match and the enforcement of the rules.

§ 5.2.1 Decisions of the Head Referee The head referee takes decisions to the best of their ability according to the rules and the spirit of the game. Any decision of the head referee is valid.

The head referee's decision is final and can not be changed afterwards, even by video proof. There is no discussion about decisions during the game, neither between the assistant referees and the head referee, nor between the audience or the teams and the head referee.

The head referee communicates their decision via verbal calls, occasionally complemented by a whistle sound.

Calls. The head referee announces decisions by a clear loud call, and (as required) whistle sound. The whistle, or where there is no whistle the first verbal word of the referees calls, defines the point in time at which the decision is made. The referees should make efforts to use consistent and clear calls, and it is preferable for referees to use the calls as specified in these rules.[4] The intention of specifying the referee calls is for clarity and consistency across games.

Whistle. Where a whistle is required, the head referee first whistles and then announces the reason for the whistle. The head referee may choose to use any normal sports whistle. Each whistle sound should be short and not too loud as to interfere with other fields and simultaneous games. The head referee must only sound the whistle in circumstances described in these rules.

§ 5.2.2 Powers and Duties The head referee enforces the rules of the game and controls the match in cooperation with the assistant referees and the GameController operator.

The head referee adjudicates all infringements of the rules, and stops, suspends, or abandons the match accordingly. They indicate the restart of play after it has been stopped.

The head referee may allow play to continue when the team against which an offense has been committed will benefit from such an advantage, and penalizes the original offense if the anticipated advantage does not actually ensue.

When a robot commits multiple offenses at the same time, only the most serious one is called.

The head referee should act on the advice of the assistant referee regarding incidents that they has not seen.

The head referee has the right to take action against team officials who fail to conduct themselves in a responsible manner and may, at their discretion, expel them from the field of play and its immediate surrounds.

The head referee must stop the game if a human is seriously injured during play until that person has left the field.

The head referee can stop, suspend, or abandon the match because of outside interference of any kind, such as but not limited to: poor lighting or network conditions, or extraneous objects, animals or people entering the field.

The head referee can request any robot be stopped if it becomes a threat to the safety of objects, participants or spectators.

The head referee ensures that no unauthorized persons enter the field of play.

The head referee should avoid handling the ball (except for placing a ball for kick-off), and avoid handling the robots. Their duty is to monitor and adjudicate the game. The head referee should only

[4]The calls specified in these rules are detailed in English. With the agreement of the teams, the referees may use suitable calls in any language. The exception to this are technical challenges that depend on the calls as specified.

handle robots and the ball if absolutely necessary to expedite game-play, where the robot handlers are otherwise occupied or too far away, and only if it doesn't compromise their own safety.

The head referee may decide at any point before or during a game to relocate any objects around the field, or direct persons to another position around the field.

§ 5.3 Assistant Referees

The assistant referees assist the head referee with offenses and other events when they have a clearer view. For example, they indicate when the ball has completely crossed over the line of the field, and which team is awarded the resulting goal or free kick. They can also point out rule violations that occurred outside of the head referee's view.

The assistant referees assist the head referee in officiating the match. They help monitor play, track the ball going out of bounds, and communicate with the head referee about incidents they observe.

Assistant referees may replace the ball when it goes off the field or becomes stuck between a player's feet. Robot handling is performed by the designated robot handlers (see § 5), not by assistant referees, unless otherwise discussed with teams beforehand.

Assistant referees should only enter the field to execute a decision made by the head referee. They should not prevent robots from falling during the game.

A game normally has one or two assistant referees. The exact number is decided by the organizers of the competition.

§ 5.4 GameController Operator

The operator of the GameController is a referee who sits at a PC in the technical area. As with the head referee, the operator should make efforts to use consistent and clear calls.

They will signal any change in the game state or penalties to the robots via the wireless as they are announced by the head referee. Note that for both kick-offs and goals, the moment of whistling is determining, not the verbal announcement of the head referee. They should repeat the call of the head referee as they do so, to confirm it was heard correctly.

The operator will also inform the robot handlers when a timed penalty is over and a robot has to be placed back on the field. They should announce events that occur automatically in the GameController due to elapsed time, such as the ball coming into play after a kick-off, penalty kick or free kick, or the state changing from `ready` to `set`.

They are also responsible for keeping the time of each half. They should count aloud the remaining seconds in a half once the time remaining is $5\,\mathrm{s}$ or less.

§ 5.5 Referee List for Friendly Games

During a competition, especially (but not only) during the setup days, several teams may want to participate in friendly games with each other if a field is available to play in. People willing to volunteer for judging these games as head referee, assistant referee or GameController operator may submit their name to a list managed by the Organizing Committee, so that the teams organizing the friendly game are aware of their availability. This is especially recommended for those who wish to gain referee experience.

Ultimately, the teams organizing the friendly game are still free to decide whether to call volunteers from the list or otherwise choose their referees.

The Organizing Committee is in charge of maintaining the referee list and should be approached at the competition site if one should want to volunteer.

§ 5.6 Team Captain

The captain is a team official. They are the representative of the team for the game. They do not have to be the team leader.

The team captain represents the team during the pre-game meeting.

In general, the team captain is the only person allowed to communicate with the referees.

Each team has one captain, who may also be a robot handler or the GameController contact (see following).

§ 5.7 Robot Handlers

The robot handlers are team officials. They are the only team members permitted to touch their team's robots, except when a robot is being serviced; in that case, any team member may touch it.

Each team must provide one or two handlers. A handler may also be the team captain, but not the GameController contact.

Robot handlers must wear a sign of identification to be easily recognized. For example, a colored armband or a specific shirt. Each team should use a different sign of identification (e.g. different colors) if possible.

During the game, the robot handlers stay in a designated area and must receive permission from the head referee before entering the field.

The robot handlers are tasked with removing their robots from the field when they are penalized, picked up, or otherwise removed from play. They must do so as swiftly as possible when instructed by the head referee, and with minimum interference with the game.

When play is stopped, the robot handlers inform the head referee when their robots are ready to resume.

Robot handlers should only enter the field to execute a decision made by the head referee. They should not prevent robots from falling during an official game.

A robot handler may not touch a robot from a different team.

Use of the Remote Emergency Button Robot handlers are allowed to carry a remote emergency stop button during a game (see § 3). This button must be worn either around the neck or on the belt of the robot handler and must be clearly marked. Robot handlers must keep their hands clearly away from the button unless the button is actively being pressed. Robot handlers can only use the remote emergency button in the following two cases: (1) command by the main referee for the robot to be stopped and removed from the game; (2) emergency situations where the robot's behavior poses danger to the humans outside of the field. Robot handlers must not use the remote emergency button to intentionally incapacitate their robots with the aim of gaining advantage in the game.

§ 5.8 GameController contact

The GameController contact is a team official. They sit next to the GameController operator and they are allowed to communicate with them, e.g. for the purpose of signaling that a penalized robot is ready to start serving its penalty time.

Each team has at most one GameController contact, who cannot also be a robot handler. The GC contact role is usually covered by the captain, unless the captain is also a handler, in which case a different team member must be designated.

§ 5.9 Humans Allowed on the Field

During the game, starting at the Ready state, only the following people are allowed on the carpeted area (i.e. on the field and the surrounding border area):

- The referees;
- The robot handlers;
- And no one else.

In addition, only the team captains and at most two more people per team are in the Technical Area (see § 1). This allows the referees easier communication with the team and the GameController operator gets less disturbed.

§ 6 Judgement

§ 6.1 Pre-game Referee Meeting and Task Delegation

At least 15 min before the start of the game, the people who are going to serve as referees meet up to discuss the upcoming game. At least, they must assign among themselves the roles of head referee and GameController operator, with the remaining people acting as assistant referees.

The head referee should also communicate whether they are going to delegate any duties to the assistants. Other topics that ensure a smooth cooperation among the referees can also be discussed.

The head referee should talk to the assistants to determine what tasks or lesser decisions, if any, they wish to delegate to them to ensure that the game is arbitrated as smoothly as possible. This is left to the discretion of the head referee, based on their expertise in the role and their ability to focus on multiple events happening in the game at the same time and apply the corresponding rules.

The head referee must clearly communicate what tasks are delegated to which person, so that everyone understands their duties during the game.

If no agreement can be found, the default is that the responsibility for most calls and decisions falls upon the head referee, as determined by the rules.

Common examples of tasks that can be delegated are:

- Indicating when the ball leaves the field
- Determining which team is to be given a free kick when the ball goes out of the field (Section 14) and communicating this to the head referee
- Indicating violations requiring a penalty, especially if they happened out of the view of the head referee

Note that the final decision is still made by the head referee.

The above list is not prescriptive: the head referee can always choose to delegate zero, one, some, or all of these tasks. It is also not exhaustive: through discussion among the head referee, the assistants, and the GameController operator, other tasks not listed here may be identified and delegated.

Care should be taken to not overburden any one person and not to blur the roles of head and assistant referees. Conflicts in the authority of the referees should be avoided, but if any do occur, the head referee's decision is final.

§ 6.2 Pre-game Team Meeting

Both teams send a representative called team captain to the field 10 min before their match starts. This time should be used to welcome each other, assign team colors, choose side and kick-off, and discuss any other topics related to the match.

§ 6.2.1 Assigning team colors. The teams agree on the jersey colors they will use during the match, following the constraints outlined in § 4. The head referee should respect the teams' wishes as long as they fit within the constraints. If a suitable agreement cannot be reached, the head referee must make the decision if possible, or propose an alternative solution.

§ 6.2.2 Choosing side and kickoff. The head referee flips a coin. The team who won the flip can decide whether they will take the kickoff or choose which side to defend during the first half. The team who lost the flip gets the other benefit.

§ 6.3 Referee – Team Communication

During the match, only the team captains are allowed to communicate with the head referee. Requests for pick-up (Section § 12.5.6) are an exception: they may be made only by the team captain or a robot handler.

Furthermore, only the team captain or the GameController contact are allowed to be in the immediate vicinity of the GC and communicate with the operator.

After the match the teams thank the referees for their duty.

During all phases of the match teams and referees are communicating with respect to each other.

§ 6.4 Referees During the Match

The head referee and the assistant referees usually oversee the game from outside the field. Robot handlers enter the field to remove or place robots when penalties are applied (see § 5). Referees should avoid interfering with the robots as much as possible.

§ 6.5 Referees After the Match

At the end of the game, the head referee must provide the Technical and Organizing Committees (or other relevant authorities) with a match report, which includes the final score and any relevant information on incidents that occurred before, during or after the match.

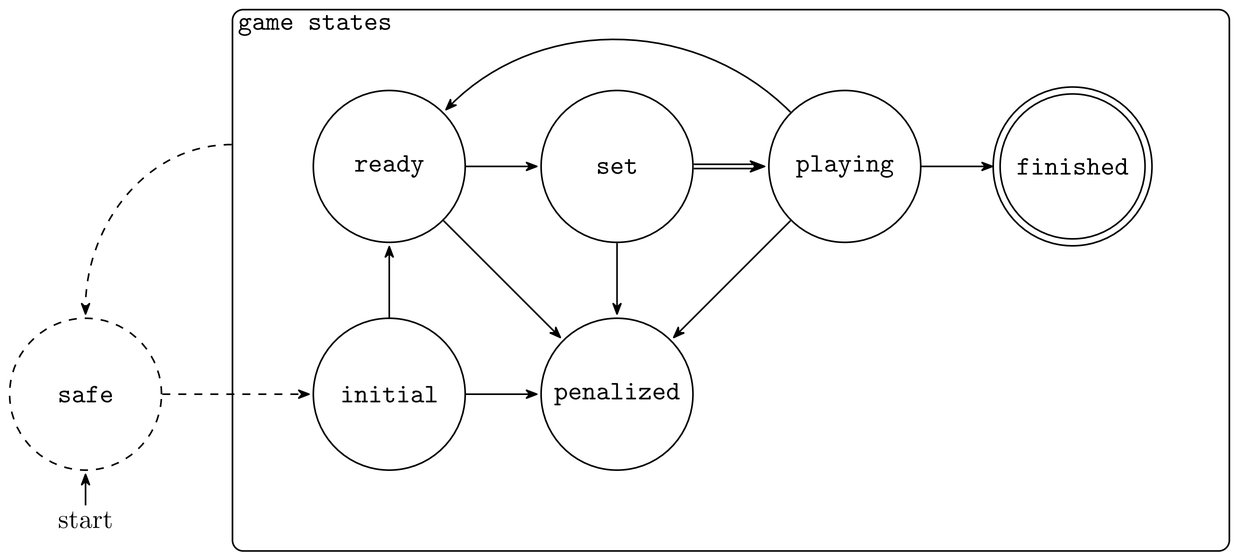


Figure 5: Diagram of the game states. Dashed lines denote technical state `safe`. Dashed transitions are activated through dedicated buttons or switches on the robot. Solid lines denote game states. Transitions between game states are activated by the GameController. The transition from `set` to `playing` (denoted by a double line) is additionally activated by a whistle. Final state `finished` is denoted by double circle.

§ 7 The Course of the Game

§ 7.1 Robot States

Throughout a match, robots transition through several defined states controlled by the GameController. Robots must respond appropriately to each state transition. Figure 5 illustrates the game states and transition between them.

§ 7.1.1 Initial The `initial` state is active before the game begins. Robots should be placed in their starting positions by robot handlers. Robots are permitted to move their heads and perform calibration routines but must not locomote (move their legs or walk).

§ 7.1.2 Ready The `ready` state begins when signaled by the GameController. During this state, robots may autonomously move to their designated positions for kick-off or other set plays. The `ready` state lasts $45\,\mathrm{s}$. Robots must reach legal positions before the `set` state begins.

§ 7.1.3 Set The `set` state indicates that robots must be stationary in their positions. Robots that could not reach legal positions by the end of `ready` are penalized for Illegal Position during this state.

No locomotion is permitted during this state, except for the minimum necessary to get up if the robot has fallen. Movement of the robot's head is also allowed. Any robot that moves its legs or locomotes during the `set` state will be penalized for Forbidden Motion (see § 12.5.2).

The head referee signals the transition from `set` to `playing` by blowing the whistle. For kick-off and drop ball situations, the GameController sends the `playing` signal with a delay of 10 s after the whistle, to encourage teams to implement whistle detection.

§ 7.1.4 Playing The `playing` state is the main game state where robots actively participate in the match. All normal game rules apply during this state.

§ 7.1.5 Penalized A robot in the `penalized` state has been removed from play due to an infringement. The robot must remain stationary outside the field until its penalty time expires. After the penalty expires, the robot autonomously re-enters the field from the designated re-entry point (see § 12).

§ 7.1.6 Finished The `finished` state indicates the end of a game half or the match. Robots should cease all game-related activity.

§ 7.1.7 Brief stop The referee may call a brief stop of play at any time. Common reasons for a brief stop are: a human needing to enter the field or approach the robot players (e.g. positioning the ball for a free kick, removing a penalized robot, ...) or emergency situations (e.g. a robot threatening to cause damage).

When a brief stop is called, all robots must safely come to a stop as quickly as possible, and then remain still and cease all motion, behaving as if in the `set` state. No locomotion is permitted, not even for getting up. Robots must respond to this signal immediately. Moving during a brief stop results in a Motion in Stop penalty (see § 12.5.3), and possibly in a caution or send-off during an emergency situation.

The head referee signals a brief stop by calling "Stop Play" loudly and clearly. The GameController will also send the stop signal.

The game clock continues to run during a brief stop, while secondary timers and penalty timers are paused. If a team was in the process of taking a kick-off or a free kick, that information is retained when the game resumes.

During a brief stop, robots remain in place and must not be moved, except for untangling. If the situation requires the robots to be removed from the field, a different call must be made, such as a referee timeout.

Play resumes when the head referee determines that the event that caused the stop is resolved. The procedure for resuming the game from a brief stop is as follows:

- The ball is placed where it was at the time of the stop (if it was in play)
- All humans leave the field
- The head referee signals play to continue
- The game resumes in the same state as it was when it was stopped.

§ 7.2 Periods of Play

The match consists of three parts: the first play period (first half), a half-time break, and a second play period (second half).

Each game period has a duration of 10 minutes, starting from the beginning of the Playing state of the initial whistle.

For the second half, the teams change sides of the field, thus attacking the goal they used to defend in the first half.

§ 7.3 Half-Time Break

The half-time break must be at least 10 minutes, unless otherwise agreed by the teams. When multiple games are played on adjacent fields, the half-time break can extend to allow the referees to move between fields, and any other time taken for such logistical considerations.

During the half-time break both teams may perform any modifications on the hardware or software of the robots, change robots, or do anything else that can be done within the time allotted.

§ 7.4 Signals

The head referee uses a *whistle* to signal certain changes in the game phases (see § 7). The head referee should make all whistle sounds from the T-junction of the halfway line. The following game phases are signaled by a whistle:

1. `playing` – one short whistle blows;
2. `finished` at the end of the first half – two short whistle blows;
3. `finished` at the end of the second half – two short plus one long whistle blow;

§ 7.5 Ready and Set States

In both knock-out and round-robin games, the game clock is not stopped during the Ready and Set states other than the initial kickoff of a period.

§ 7.6 Allowance For Time Lost

Allowance is made in either period for all time lost through:

- substitutions
- assessment of injury to players
- removal of injured players from the field of play for treatment
- wasting time
- external circumstances (e.g., network issues)
- any other cause

The allowance for time lost is at the discretion of the head referee. It is announced during the final minute of each half, and it is always expressed in whole minutes. The GameController operator executes this addition of time using the GameController interface.

In exceptional circumstances, the head referee may decide to add time to the clock immediately after an incident rather than waiting until the final minute of a half.

§ 7.7 Timeout

§ 7.7.1 Team Timeout Each team can call a **maximum of 1 timeout per game** with a total time of no more than **5 minutes**. During this time, both teams may change robots, change programs, or anything else that can be done within the time allotted.

A team may only call their timeout at one of these times:

- Before a period of play begins (including a half of extra time)
- During the `ready` and `set` states pertaining to a stoppage of play that is restarted with a kick-off or a dropped ball (see § 8, § 8), for example after a goal or a Global Game Stuck event (see § 8)
- Before a penalty shootout (see § 14)

The timeout ends when the team that called the timeout says they are finished, at which time they must be ready to play. The other team must be ready to play at the time the timeout runs out, or **2 minutes** after a prematurely called end of the timeout, whichever is earlier. If the other team is not ready to play in time, it must call a timeout of its own.

The clock stops during timeouts and is reset to the time when the current stoppage of play began.

After the completion of the timeout, the game resumes with a kick-off for the team which did not call the timeout.

If a team is not ready to play at the assigned time, it must use its timeout. After the timeout has expired, or if the team does not use it, the referee shall start the game with only the ready team on the field, if the other team is still not prepared. The unready team may return its robots to the field by having a robot handler place them on the standard re-entry point (see § 12) and informing the GameController operator. The robots will join the game immediately. The team may return all its robots to the field at the same time, or return different robots (or groups of robots) at different times.

If both teams are not ready, the same procedure applies.

§ 7.7.2 Referee Timeout The head referee may call a timeout at any time if deemed necessary. A referee timeout should only be called in exceptional circumstances—one example might be when the wireless network is down or no robot responds to the GameController.

Referees may call multiple timeouts during a game if needed. Teams may do anything during these timeouts, but they must be ready to play **2 minutes** after the referee ends a timeout. The referee should end the timeout once the circumstance for which the timeout was called has been resolved. In cases where the circumstance is not resolved within 10 minutes, the Technical Committee should be consulted regarding when or if play should continue.

If a team was going to have kick-off at the time the referee timeout was called (e.g., during most stoppages of play), that team shall kick-off when the game resumes. Otherwise, the drop ball rule (see § 8) applies instead.

§ 7.8 Mercy Rule

A game will conclude once the game score shows a goal difference of 10. Ending the game is mandatory once a goal difference of 10 is reached.

§ 7.9 Ball Stop Rule

If the time for a period would end while the ball is still in motion, the period is extended instead until any of the following events happens:

- The ball comes to a complete stop, or

- The ball leaves the field of play. If this causes a valid goal to be scored, the goal is counted.

This extension should last a few seconds at most.

§ 7.10 Penalty Kick Extension Rule

If the time for a period would end while a penalty kick is taking place, the period is extended instead until the penalty kick is completed (see § 14). Note that this rule applies to any game state as long as the penalty kick substate is active, including the Ready and Set preceding the actual kick.

§ 8 The Start and Restart of Play

A kick-off starts both halves of a match, both halves of extra time and restarts play after a goal has been scored. Free kicks (direct or indirect), penalty kicks, throw-ins, a goal kicks and corner kicks are other restarts (see Laws 13–17). A dropped ball is the restart when the referee stops play and the Law does not require one of the above restarts.

If an offense occurs when the ball is not in play, this does not change how play is restarted.

§ 8.1 Kick-off

§ 8.1.1 Procedure The team taking a kick-off is determined as follows:

- The initial kick-off of the first half is taken by the team who got this benefit during the pre-game meeting (see 6).
- The initial kick-off of the second half is taken by the team who did not take the initial kick-off of the first half.
- After a team scores a goal, a kick-off has to be taken by their opponents.
- Other situations, such as timeouts, have their own rules for determining the kick-off team.

For every kick-off, during the **set** state, the ball is positioned on the center mark and is stationary. The players must follow these positioning rules:

- all players of the team *not* taking the kickoff must be in their own half of the field of play and outside of the center circle,
- all players of the team taking the kickoff must be in their own half of the field or inside the center circle,

- at most one player of the kicking team may be outside its own half (it must still be inside the center circle). That player is designated as the kicking player for this kickoff.

Robots that do not follow these rules are penalized for illegal position. Then, the head referee gives the signal to start play with a whistle. At this point, the following rules apply:

- the positioning rules above still apply until the ball is in play,
- if a kicking player was designated during the **set** state, that player *must* take the kickoff. If a different player kicks the ball before the ball is in play, this is an offense and the kick-off is retaken,
- the ball is in play when it is kicked and clearly moves, or 10 s after the referee has given the signal, whichever happens first. In either case, the head referee or the GameController operator will verbally announce this by the call "Ball Free",
- a goal may **not** be scored directly against the opponents from the kick-off,
- if the team taking the kick-off has three or more robots on the field then two different robots need to touch the ball before scoring a goal. If a team taking the kick-off has only two or less robots on the field, the robot taking the kick-off has to touch the ball at least one time outside the center circle before scoring a goal.

In order to encourage teams to recognize the signal to start play, the GameController application will delay the **playing** signal, as specified in § 7.1.3. The delay applies regardless of when or how the ball becomes in play.

§ 8.1.2 Offenses and Sanctions

1. The regulations defined in the section § 12 apply (this includes in particular § 12.5.2 and § 12.5.1).
2. In the event of any offense during the kick-off procedure, the kick-off is retaken.

§ 8.2 Dropped Ball

§ 8.2.1 Definition A *dropped ball* is a method of restarting play when the referee stops the game while the ball is in play, and no team is entitled to a kick-off, free kick, or any other specific restart under these Laws.

Typical situations include, but are not limited to:

1. referee or system error,
2. a pause in game during the `playing` state that has been acknowledged by the head referee,
3. a global game stuck (see § 8.3.2).

§ 8.2.2 Procedure

1. **Announcement and Preparation:** The head referee announces “Dropped Ball”. The game state switches to `ready` phase. No team is designated to take kick-off.
2. **Ready:** Robots assume their positions inside their own half.
3. **Set:** The robots must stop moving. No robot of either team is allowed inside the center circle. The referee places the ball on the center mark.
4. **Playing:** The referee signals the start and the game switches to `playing` phase (see § 7). The ball is immediately in play. A goal may be scored directly from a dropped ball.

§ 8.2.3 Offenses and Sanctions

1. The regulations defined in the section § 12 apply (this includes in particular § 12.5.2 and § 12.5.1).
2. In the event of any offense during the dropped ball procedure, the dropped ball is retaken.

§ 8.2.4 Notes A dropped ball follows the standard kick-off procedure § 8 (including `ready` and `set`), with the following exceptions: (1) no team has kick-off rights; (2) no robot is allowed inside the center circle before the `playing` state; (3) the ball is immediately in play; (4) a goal may be scored directly from a dropped ball.

§ 8.3 Game Stuck

Game stuck situations occur when play has stalled and intervention is required to resume normal game flow.

§ 8.3.1 Local Game Stuck Local game stuck is called when the ball has not moved significantly for 10 s while robots are in proximity to the ball. This typically occurs when robots are clustered around the ball but unable to make progress.

When local game stuck is called:

- The robot nearest to the ball is penalized and removed according to the standard removal penalty.
- The ball remains in its current position.
- Play continues immediately after the robot is removed.

The head referee calls “Game Stuck <robot color><robot number>”.

Local game stuck penalties do not count towards the incremental penalty count.

§ 8.3.2 Global Game Stuck Global game stuck is called when no robot has been within $1\,\mathrm{m}$ of the ball for $30\,\mathrm{s}$. This typically occurs when all robots have lost track of the ball or are unable to approach it.

When global game stuck is called:

- Play is stopped.
- The game is restarted with a drop ball (see § 8).

The head referee calls “Global Game Stuck”.

§ 9 The Ball In and Out of Play

The definition of the ball in/out of play refers to the definition of *inside* and *outside* in Section 1 (Definition of Inside and Outside).

§ 9.1 Ball Out of Play

The ball is out of play when:

- the ball is outside the field of play (i.e. not even touching/overlapping the boundary lines)
- the ball is inside the goal
- the ball is being handled by a referee during play (such as the referee replacing the ball under Section 4 (Judgment))
- play has been stopped by the referee

§ 9.2 Ball In Play

The ball is in play at all other times, including (but not limited to) when:

- the ball rebounds off the head referee or a assistant referee, the goalpost, or the crossbar, and remains inside the field of play (i.e., within the region bounded by the touchlines and goal lines)
- the ball rebounds off a robot player, including a robot located beside the field but outside the field of play, provided that the ball has not left the field of play during the rebound
- the ball rebounds off a robot handler who has been permitted by the head referee to enter the field of play, provided that the head referee determines that the contact was not intentionally caused by the robot handler

§ 10 The Method of Scoring

§ 10.1 Goal Scored

A goal is scored when the entire ball passes over the goal line, between the goalposts and under the crossbar as defined in Section 1 (Definition of Inside and Outside), provided that no infringement of the Laws of the Game has been committed previously by the team scoring the goal.

The restart of the play will be a kick-off for the opponents team.

§ 10.2 Winning Team

The team scoring the greater number of goals during a match is the winner. If both teams score an equal number of goals, or if no goals are scored, the match is drawn.

§ 10.3 Permitted Tiebreakers

When competition rules require there to be a winning team after a match or home-and-away tie, the only permitted procedures for determining the winning team are:

- Extra time (2 additional equal periods of no more than 5 minutes each)
- Penalty shoot-out (see § 14)

Before each additional time period and penalty shoot-out, a five-minute break shall take place. This break follows the same rules as the half-time break (see § 7).

§ 11 Offsides

Offsides are not used in the RoboCup Humanoid Soccer League.

§ 12 Fouls and Misconduct

Actions that break the rules of play are penalized. The referee should use time penalties as the primary method of enforcing routine gameplay infractions and minor fouls (see § 12). However, cautions and send-offs are available at the referee's discretion for serious, dangerous or repeated misconduct (see § 12).

When a robot commits an infraction, the head referee shall call out the **infraction** committed, the **team color** of the robot, and the **jersey number** of the robot. If available, the referee also raises the appropriate card to indicate the type of penalty that is applied. A blue card is used for time penalties, a yellow card is used for cautions, and a red card is used for send-offs.

The penalty for the infraction will be applied immediately. Only the robot handlers of the penalized team should carry out the movements required to serve the penalty, without involvement from referees or any other member of either team. The operator of the GameController will send the appropriate signal to the robots indicating the infraction committed.

If a robot commits an offense in the vicinity of the ball (within the free kick avoidance radius, which is the center circle radius for the respective field size - see Table 2), the game is interrupted and the offending player is penalized. After the penalty has been applied, the opposing team restarts play with a free kick (see § 13). If the offense occurred away from the ball, the offending robot is penalized, but play is not interrupted.

§ 12.1 In-Place Penalty

One infraction in particular has the robot penalized *in-place*. Such a robot is not removed from the field and is not handed to the team for maintenance, but rather, it remains where it is on the field and immediately starts serving its penalty time.

A robot penalized in-place must safely come to a stop immediately, and then is not allowed to move in any fashion until its penalty time expires. As an exception, the penalized robot is allowed to attempt to get up as normal.

The time for an in-place penalty is always $15\,\mathrm{s}$.

In-place penalties do not increment the duration of subsequent standard removal penalties.

If the penalized robot moved significantly from the position where it committed the infraction before stopping, the head referee may require a robot handler to reposition the robot to the point of the infraction, potentially calling a brief stop if necessary.

In-place penalties are considered to be over at the end of each half.

§ 12.2 Standard Removal Penalty

Robots receiving a *standard removal penalty* are removed from the field of play for a specified amount of time, after which they are returned to the field of play.

When the head referee indicates an infraction has been committed that results in the standard removal penalty, the robot handler for that team will remove the robot immediately from the field of play. The robot should be removed in such a way as to minimize the movement of the other robots and the ball. If the ball is inadvertently moved when removing the robot, the ball should be replaced to the position it was in when the robot was removed.

The GameController will send the appropriate penalty signal to the robot indicating the infraction committed. After a penalty is signaled to the robot, it is not allowed to move in any fashion except for standing up or sitting down.

The initial duration of the standard removal penalty is $45\,\mathrm{s}$. Unless otherwise specified, each additional offense committed by a player on the same team increases that team's penalty duration by $10\,\mathrm{s}$. Penalty durations are tracked separately for each team. During the `set` state the penalty time counter will not decrease. The GameController will keep track of the time of the penalty. To avoid disproportionately penalizing minor infractions, the infractions *Illegal Positioning*, *Incapable Robot* and *Local Game Stuck* result in a standard removal penalty but do not increase the penalty timer.

Example: Player red 1 pushes player blue 1. As it is team red's first offense of the game, player red 1 is removed from the game for $45\,\mathrm{s}$ and team red's penalty tracker increases by $10\,\mathrm{s}$. Player red 2 now commits another offense by leaving the field. Player red 2 is removed from the game for $55\,\mathrm{s}$ and their penalty tracker again increases by $10\,\mathrm{s}$. Finally, player blue 1 pushes player red 3. As team blue has not yet committed an offense, blue 1 is removed from play for $45\,\mathrm{s}$ and team blue's penalty tracker increases by $10\,\mathrm{s}$.

After a robot is penalized and before it starts serving its penalty, the robot handler may hand it over to its team who may perform maintenance on the robot. The penalty timer does not decrease while maintenance is being performed.

To serve its penalty, the removed robot is placed on the touchline at the height of the penalty mark of its own half and facing the opposite touchline. The robot must be placed fully outside of the field. If the position at the height of the penalty mark is already occupied, a nearby position is chosen. When finding a nearby location, it **must** still be in the robot's own half, so that the symmetry of the field can be resolved by the robot's localization system. After the robot is placed, the robot handler announces "<robot color><robot number>Ready", after which the GameController operator starts the penalty timer. After the penalty time elapses, the GameController operator will remove the penalty from the penalized robot, enabling it to autonomously re-enter the game without requiring additional intervention from the referees.

If any team member interferes with the robot after it has started serving its penalty time (including button presses or connecting cables), the team member receives a warning, and the penalty timer is reset.

Standard removal penalties are considered to be over at the end of each half.

§ 12.3 Cautions and Send-Offs

Referees may issue cautions or send-offs at their discretion for serious fouls, dangerous play or unsportsmanlike conduct. Routine gameplay infractions should be managed through time penalties instead. A caution results in a time penalty in addition to the caution itself. If a robot that has already received a caution is cautioned again, it is sent off. In especially serious cases, the referee may issue a direct send-off without first issuing a caution.

When a caution or send-off is issued, the referee shall call out ”Caution/Send-Off <robot color><robot number>”. The same removal procedures as for the standard removal penalty apply.

Cautions are assigned to individual robots and remain associated with that robot for the duration of the match, regardless of substitutions. A substituted robot retains any cautions it has received and will still be considered cautioned if it later re-enters the match. A robot replacing a cautioned robot does not inherit its caution status.

A send-off permanently removes the robot from the match. A sent-off robot may not be replaced by a substitute, resulting in the team continuing the match with one fewer robot.

Cautions and send-offs issued against robots are cleared at the end of the game.

Offenses that can result in a caution or send-off include, but aren’t limited to

- Damage to the field
- Damage to other robots
- Endangering any human
- Repeated deliberate violations of the rules

It is at the discretion of the head referee to decide whether a caution or send-off is warranted.

Teams are ultimately responsible for their robots to play in a safe and sportsmanlike manner. Repeated, *willing* dangerous behavior (including neglecting to correct a safety issue discovered previosuly), especially over multiple games, may be considered unsporting conduct by the team (see below).

§ 12.4 Penalties Against Teams or Humans

The referee can issue warnings, yellow and red cards against humans or teams at their discretion. Cards can be issued for offenses such as:

- Offensive, violent or abusive behavior
- Repeatedly ignoring referee instructions
- Repeatedly entering the field without permission
- Breaking the game rules or exploiting loopholes to gain an unfair advantage
- Repeated, willing dangerous behavior by the team's robots (see above)

Cards that are issued against humans or teams must be reported to the Organizing Committee immediately after the game. Cards against humans or teams persist between games. A human who receives a red card will be prohibited from entering the field or team area during games for the remainder of the competition. A team that commits serious or repeated violations may be disqualified from the competition.

§ 12.5 Gameplay Infractions

This section describes infractions commonly encountered during regular gameplay.

§ 12.5.1 Illegal Positioning A robot is considered to be illegally positioned if, during the `set` state, it is located outside the field of play, or violates the positioning rules for the current phase of play (see Section 8 for kick-off and 13 for penalty kicks). A robot is also considered to be illegally positioned if it encroaches into the avoidance areas during a free kick (see 13) or penalty kick (see 13). The referee should not measure exact distances for avoidance areas and shall only penalize robots that are clearly violating this rule. A robot making a reasonable effort to return to the field of play following a standard penalty shall not be penalized for illegal position.

Illegally positioned robots are subject to the standard removal penalty (see § 12). The head referee will call "Illegal Position <robot color><robot number>". Referees may use "Illegal Defender" interchangeably to help with clarity. For simplicity, Illegal Positioning penalties do not count towards the incremental penalty count.

During the `set` and `playing` states, at most *three* players can be within their own goal area at the same time. A robot is within the goal area if any part of its body is touching the ground inside the goal area or touching one of its lines. An "Illegal Position" penalty is applied when any additional players enter the area in `playing`, or to the excessive players closest to the border of the goal area in `set`. Note that if a player is pushed into the goal area by an opponent, this robot will not be subject to removal, unless it fails to exit the area within 5 s (or 5 s after getting up if the pushing led to falling).

If an illegal defender kicks an own goal, the goal is scored for the opponent. If there is any doubt about whether a goal should count (e.g., the illegal defender infraction is called, but the robot scores the own goal immediately afterwards, before it is removed), then the decision shall be against the infringing robot.

§ 12.5.2 Motion in Set Robots that begin moving their legs or locomote in any way during the `Set` phase, except for standing up after a fall, are penalized in-place. The head referee will call "Motion in Set <robot color><robot number>". Note that falsely responding to a whistle on another field will result in this penalty. This rule does not apply to the movement of heads or arms. Motion in Set penalties do not count towards the incremental penalty count.

§ 12.5.3 Motion in Stop Robots that begin moving their legs or locomote in any way during a brief stop § 7.1.7, are awarded a standard penalty. The head referee will call "Motion in Stop <robot color><robot number>".

§ 12.5.4 Local Game Stuck Penalty When Local Game Stuck is called (see § 8.3.1), the nearest robot to the ball will be penalized according to the standard penalty. Local Game Stuck penalties do not count towards the incremental penalty count.

§ 12.5.5 Incapable Robot A robot is considered incapable, if it has ceased activity for 10 s or has been turned off.

For the purpose of this rule, a robot is considered active, if it performs at least one of the following actions:

1. Walking in any direction or turning
2. Searching for the ball or tracking the ball

A robot that has fallen is considered incapable, if it does not attempt to get up within 20 s or fails to get up within 3 attempts, or 4 when externally disturbed during initial attempts.

An incapable robot must be removed from the field of play and awarded a standard penalty. In such cases, the head referee shall announce "Incapable robot <robot color><robot number>". Incapable Robot penalties do not count towards the incremental penalty count.

Note: This rule is not intended to penalize robots solely for remaining stationary, provided they are responsive and have not crashed.

Player Stance. Robots must not maintain a stance wider than a reasonable proportion of their body width for longer than $10\,\mathrm{s}$. As a guideline, the stance should not exceed approximately 1.5 times the robot's shoulder width. A robot staying in an excessively wide stance for longer than $10\,\mathrm{s}$ is considered to be incapable of playing, resulting in a standard removal penalty (see above).

§ 12.5.6 Request for Pick-up A robot handler or team captain may request to pick up a robot if and only if a robot is in a dangerous situation that is likely to lead to physical injuries. The head referee must approve this request prior to the robot handler touching the robot. If a team member touches a robot without the permission of the referee, the robot receives a standard removal penalty and the person touching it receives an official warning.

The referee may deny a request for pick-up. A pick-up is not granted for reasons such as being incorrectly configured or being unable to see the ball.

§ 12.5.7 Ball Holding The goalkeeper is allowed to hold the ball for up to $10\,\mathrm{s}$ as long as it has at least one foot inside its own penalty area. In all other cases, robots are allowed to hold the ball for up to $5\,\mathrm{s}$. Holding the ball for longer than this is not allowed. A robot is holding the ball, if the convex hull of the robot is covering more than half of the ball. It does not matter whether the offending robot is actually touching the ball. Repeatedly releasing and holding the ball in quick succession is treated as one continuous hold, even if individual holds are shorter than the limit. In general, robots should release the ball for approximately as long as they were holding it to reset the clock.

Ball holding may not be called when a robot falls on a ball. The robot will either get up and hence free the ball, or the robot should be removed under the Fallen Robot rule. Further, ball holding may not be called when the ball becomes stuck between a robot's legs. In such a situation, the head referee should call "Clear Ball" and a robot handler or assistant referee should remove the ball and place it approximately where it was before it became stuck.

If a ball holding offense was committed, the head referee will call "Ball Holding <robot color><robot number>", and the robot is removed under the standard removal penalty. The ball should be removed from the possession of the robot and placed where the penalty occurred. If the robot that held the ball has moved the ball before the robot can be removed, the ball shall be replaced where the penalty occurred. This also applies to accidental goals.

Example: A robot holds the ball, and before the referees can remove the robot, it shoots the ball into the goal. The goal will not be counted, and the ball will be replaced where the penalty occurred.

§ 12.5.8 Leaving the Field A robot that leaves the carpeted playing area will be subject to the standard removal penalty. The head referee will call "Leaving the Field <robot color><robot number>".

Additionally, a robot will also be subject to the standard removal penalty when:

- The robot walks into the goalposts or goal net for more than 5 s, including robots that are stuck on the goalposts and unable to free themselves.
- The robot's fingers or other appendages become entangled in the net (without any time constraint).

§ 12.5.9 Playing with Arms or Hands Playing with arms/hands occurs when a field player or a goalkeeper outside its own penalty area moves its arms/hands to touch the ball (except during a fall or get-up). The goalkeeper is allowed to touch the ball with its arms/hands while it is within its own penalty area.

A robot playing with arms/hands will be subject to the standard removal penalty. The ball will be replaced at the point where it contacted the arms/hands of the offending robot. If an own goal is scored as a result of playing with arms/hands, the goal should count and the player should not be penalized.

Accidental playing with arms/hands when a robot falls or executes a get-up routine will not be penalized. If the ball goes out of play in this case, normal kick-in rules will apply (see § 15). However, goals (except for own goals) resulting from a ball contact with the arms/hands during a fall or get-up do not count and result in a Goal Kick (see § 16) as if the ball went over the goal line next to the goal.

§ 12.5.10 Pushing Pushing is any direct or indirect contact with an opponent robot that is either forceful enough to destabilize the opponent or continues for more than 5 s, even when the applied force is minimal. Front-to-front contact between robots does not constitute pushing, unless one robot is moving with significantly more force or speed than the other. Pushing may only occur between players of different teams. A robot pushed by another robot cannot simultaneously be called for pushing itself. In this case, only the initially offending robot will be penalized. A stationary robot cannot be penalized for pushing, including a robot that is kicking, provided that the ball was close enough where a kick could have succeeded at the start of the kick motion.

If a pushing offense is committed, the head referee will call "Pushing <robot color><robot number>" and the offending player will be removed from play according to a standard penalty. If the ball moves significantly as a result of pushing, then it should be replaced to where it was at the time of the infraction.

§ 12.6 Manual Interaction by Team Members

Manual interaction with the robots, either directly or via some communications mechanism, is not permitted. Team members (that are not robot handlers) can only touch one of their robots when a robot handler hands it over to them after a it has been penalized or substituted.

§ 12.7 Aborting a Penalty

If a penalty was incorrectly applied to a robot and acknowledged by the referee to be incorrect, the robot should first be removed and placed outside the field, following the same process as for the Standard Removal Penalty (see § 12). Once placed, the robot should then be allowed to continue play immediately after being placed and the GameController removes its penalty.

The aborted penalty will not be added to the total team penalty count and will not contribute to the incremental penalty increase.

§ 12.8 Untangling

If entangled robots fail to untangle themselves, the referee may ask designated robot handlers of both teams to untangle the robots. Untangling must not make significant changes to robot positions or heading directions. Untangled robots must be placed on the ground no closer than $50\,\mathrm{cm}$ to the ball and in a way that does not give either team an advantage. Being entangled does not incur a penalty.

§ 12.9 Jamming

During a match, robots shall not jam the communication and sensor systems of the opponents:

§ 12.9.1 Wireless Communication If a team violates the general regulations defined in § 3, penalties will apply. First violation will result in a warning. Repeated violation might result in disqualification from the tournament (including all technical challenges and side competitions).

§ 12.9.2 Wireless Robot-To-Robot Communication Any team that exceeds the limits for wireless robot-to-robot communication defined in § 3.9.3, as tracked by the GameController, will have all goals automatically nullified for the game.

§ 12.9.3 Acoustic interference Both teams and the audience shall avoid intentionally confusing the robots by producing sounds similar to the game whistle or to sounds used by the robots to communicate with each other.

§ 12.9.4 Visual perception The use of flashlights or other bright light sources is not allowed during games. However, flash photography from the audience is allowable as long as the head referee believes the purpose is not to jam any of the robots.

§ 13 Free Kicks

A free kick is a method of restarting play after various stoppages.

A free kick is initiated in the following situations:

- When the ball goes over the touchlines, termed *Kick-in* or *Throw-in* (see § 15).
- When the ball goes over the goal line having last touched a player of the defending team, termed *Corner Kick* (see § 17).
- When the ball goes over the goal line having last touched a player of the attacking team, termed *Goal Kick* (see § 16).
- When an infringement is awarded that requires play to be stopped (see § 12), termed a *Pushing Free Kick*.

The head referee will announce a free kick by calling one of:

1. “Kick-in <color>” / “Throw-in <color>” for the team awarded the kick-in.
2. “Corner Kick <color>” for the team awarded the corner kick.
3. “Goal Kick <color>” for the team awarded the goal kick.
4. “Foul <offending color><offending number>” for pushing free kicks.

The GameController will then activate the sub-state for the respective free kick. The team awarded the free kick (termed the *attacking team*) has 45 s to complete the kick. Free kicks are classified as either **direct** or **indirect** (see § 13).

§ 13.1 Direct Free Kick

A direct free kick allows the attacking team to score a goal directly from the kick:

- If a direct free kick is kicked directly into the opponent's goal by the attacking team, a goal is awarded.
- If a direct free kick is kicked directly into the team's own goal, a corner kick is awarded to the opposing team.

The following set plays are direct free kicks: *Corner Kick*, *Goal Kick*, and *Pushing Free Kick*.

Note: If the attacking team fails to execute the free kick within the time limit, the defending team may also score directly (see Failed Free Kick in § 13).

§ 13.2 Indirect Free Kick

An indirect free kick requires the ball to be touched by another player before the attacking team can score a goal:

- A goal can only be scored by the attacking team if the ball has been touched by another player (of either team) after the kick, before being kicked again into the goal.
- If an indirect free kick is kicked directly into the opponent's goal by the attacking team without touching another player, a goal kick is awarded to the opposing team.
- If an indirect free kick is kicked directly into the team's own goal, a corner kick is awarded to the opposing team.

The following set plays are indirect free kicks: *Kick-in* / *Throw-in*.

Note: If the attacking team fails to execute the free kick within the time limit, the defending team may score directly without the indirect requirement (see Failed Free Kick in § 13).

§ 13.3 Execution

The referee places the ball according to the type of free kick (see § 15, 16 and 17 for specific ball placement rules). For a *Pushing Free Kick*, the ball is left in place.

§ 13.3.1 Ball Position and Placement All free kicks are taken from the place where the offense occurred, except:

- Indirect free kicks to the attacking team for an offense inside the opponents' penalty area are taken from the nearest point on the penalty area line which runs parallel to the goal line.
- Free kicks to the defending team in their goal area may be taken from anywhere in that area.

When the referee calls for a free kick, play must also be stopped (see § 7.1.7). Once the robots become stationary, the referees will place the ball in the appropriate position. Then, the head referee gives the signal to resume play and commence the free kick.

The ball must be stationary as play resumes, and is in play when it is kicked and clearly moves as determined by the referee. The ball is also considered in play 45 s after the referee gave the signal.

§ 13.3.2 Avoidance Region and Distance Requirements During a free kick, only the attacking team may approach within the avoidance region of the ball. For all free kicks, the avoidance region is the center circle radius for the respective field size (see Table 2). This means the same exclusion zone as during kick-off applies to all free kicks, an opponent robot must remain outside this radius from the ball until it is in play.

Until the ball is in play, all opponents must remain outside of this avoidance region, unless they are on their own goal line between the goalposts, and outside the penalty area for free kicks inside the opponents' penalty area. Any players that find themselves in violation of these restrictions as the free kick is awarded must immediately make an effort to move away from the ball to outside the avoidance region until the violation is corrected.

Defensive robots that violate this rule are penalized with "Illegal Positioning" (see § 12.5.1), which results in a standard removal penalty (see § 12). Additional penalties against any further robots during the free kick, including pushing, do not result in an additional free kick but still incur the appropriate removal penalty.

§ 13.3.3 Sanctions and Retakes If, when a free kick is taken, an opponent is violating the rules above, that player receives a removal penalty and the kick is retaken.

Note that, in particular, a defensive robot that had found itself already inside the avoidance zone as the free kick was called cannot be penalized if the kick is taken before it had time to leave, as long as it had been making an effort to move away from the ball.

§ 13.3.4 Completion A free kick is deemed completed and play returns to normal when:

- The attacking team moves the ball clearly (except for a robot getting up, which is exempt from this rule), or
- The 45 s time period expires (or game time expires).

The head referee announces completion by calling “Ball Free”, and the GameController resumes the `playing` state.

§ 13.3.5 Failed Free Kick If the attacking team fails to execute the free kick within the allotted 45 s, the free kick expires and play resumes normally. In this situation:

- The defending team may play the ball immediately after the free kick expires.
- If the defending team plays the ball first (before the originally attacking team), they may score a direct goal regardless of whether the original free kick was direct or indirect.

Example: The blue team is awarded a kick-in (indirect free kick) but does not play the ball within 45 s. The referee calls “Ball Free”. A red robot immediately kicks the ball directly into the blue goal. The goal counts, as the failed-free-kick exception allows the defending team to score directly.

§ 14 Penalty Kick and Penalty Shoot-Out

§ 14.1 Penalty Kick

A penalty kick is awarded against a team that commits an offense for which a **direct free kick** (see § 13) is awarded, inside its own penalty area and while the ball is in play. A goal may be scored directly from a penalty kick.

§ 14.1.1 Procedure During normal play, if a penalty kick is awarded, the head referee calls “Penalty Kick <color>” for the team that will be performing the kick[5]. The GameController immediately transitions to the `ready` state, with a specific field in the message informing the robots that a penalty kick is about to take place.

During the `ready` state, the robots have 30 s to reach valid positions for the penalty kick, according to the following rules:

- The defending goalkeeper must have both its feet on the goal line.
- At most one robot of the kicking team may be inside the defending team’s penalty area. It may not block the penalty mark during the `set` state.
- All other robots must be outside the penalty area and at least one center circle radius (see Table 2) from the penalty mark.

[5] This is a change from previous Standard Platform League (SPL) rules.

- If multiple robots of the kicking team violate these rules, the only robot to not be penalized shall be the one that is closest to the penalty mark without blocking it.
- All robots other than the defending goalkeeper must be behind the defending team's penalty mark.

When the `ready` time ends, the GameController transitions to `set` with the penalty kick sub-state active. The following actions take place:

- If the defending goalkeeper is not positioned correctly, it is manually placed on the closest point of the goal line that is not blocked by a goal post. The goalkeeper is moved in a direction perpendicular to the goal line, and its orientation is left unchanged.
- The positioning rules above enter into effect. Robots that violate them are penalized for illegal position.
- The referees position the ball on the penalty mark.

The head referee commences the penalty kick by blowing the whistle once and calling "Playing". The GameController activates the `playing` state after the standard delay (see § 7.1.3). During this state:

- The defending goalkeeper must remain on its feet (it is not allowed to "dive") until the ball is touched by the kicking team or the penalty kick is complete.
- The positioning rules remain into effect for the same duration.

The time limit for the penalty kick is $60\,\mathrm{s}$ after play begins. The penalty kick ends when:

- The ball comes to a complete stop after the first contact by the striker robot.
- The striker robot touches the ball a second time after the ball has clearly moved.
- The ball leaves the field of play.
- The time limit expires.
- A goal is scored.

A goal is awarded to the attacking team if the ball completely crosses the goal line between the goalposts and under the crossbar before the penalty kick ends, provided no infringement has occurred.

§ 14.1.2 Infringements and Sanctions If the goalkeeper violates the positioning rules (leaves the goal line early or dives before the striker touches the ball), the penalty kick ends immediately and a goal is awarded to the attacking team.

If the striker violates the rules (touches the ball a second time after it has moved), the penalty kick ends immediately with no goal awarded.

All standard rules such as Ball Holding and Pushing are applied during the penalty kick, but they do not result in additional free kicks. A goalkeeper will not be penalized for inactivity during a penalty kick, provided it is actively attempting to defend the goal.

§ 14.2 Penalty Kick Shoot-Out

A penalty kick shoot-out is used to determine the outcome of a tied game when an outcome is required (for example, during knockout rounds, quarterfinals, semifinals, third place match, or final).

All penalty kicks are taken against the same goal.[6]

The team listed first on the competition schedule will have the striker robot for the first penalty kick. Subsequently, both teams take kicks alternately. The penalty kick shoot-out will consist of three penalty kicks per team. A team that has scored the most goals at the conclusion of these will be declared the winner. A winner can also be declared before the conclusion of the penalty shoot-out if a team can no longer win mathematically. If the two teams remain tied after three penalty kicks, then a sudden death shoot-out will follow until a definite winner is found.

The penalty kick shoot-out starts immediately without changing any code after the second half ended. No timeouts may be called during the penalty shoot-out. However, a team may request a timeout before the penalty shoot-out starts if they have a timeout remaining for this game. During the timeout, code changes are allowed.

Before the penalty shoot-out begins, each team must hand over to referees up to 6 prepared robots that may participate in the penalty shoot-out. No robots may be added once the penalty shoot-out starts. Robots that will not participate in the shoot-out must not be on the wireless network and must stay outside of the field. All participating robots must be wearing the correct jersey for their player number and no duplicate numbers are permitted.

During the entire penalty kick shoot-out, the robots are controlled by the GameController and the referee's whistle. A team cannot request the referees to press buttons on their robots, except for initially leaving the `initial` state.

§ 14.2.1 Penalty Kick Procedure A penalty kick is carried out with one striker robot and one opposing goalkeeper. Before each penalty kick, both teams must select the robot to participate (as goalkeeper or striker). The team leader communicates the selection to the head referee by

[6] Which goal to use for the shoot-out is decided in agreement with the teams, or otherwise by a coin toss.

privately handing the referee a card with their chosen number. After both teams have selected their player, the GameController operator selects the requested striker and goalkeeper robots and the GameController communicates that all non-selected robots are substitutes and should remain inactive. The GameController indicates which team has the striker robot in the current penalty kick.

Referees place the ball, the striker, and goalkeeper robots. The ball is placed on the penalty mark closest to the goal being defended. The striker robot is positioned on the edge of the penalty area, facing the ball and the goal. The goalkeeper is placed with its feet on the goal line and in the center of the goal. Neither robot is permitted to locomote during the `set` state.

The head referee commences the penalty kick by blowing the whistle once and calling "Playing". The GameController activates the `playing` state after the standard delay.

The time limit for the striker is $60\,\mathrm{s}$ after the penalty kick starts. A penalty kick is over when the ball has come to a full stop after the first contact by the striker robot. The striker robot is not allowed to play the ball a second time after the ball has clearly moved, otherwise the penalty kick ends immediately. A goal is awarded to the attacking team if a goal has been scored (the ball has completely crossed the goal line) before the penalty kick is over. Otherwise, the score is unchanged.

The goalkeeper robot must always be in contact with the goal line and must remain on its feet until the striker robot touches the ball. The goalkeeper is only permitted to "dive" and be off its feet after the attacking robot has touched the ball. If the goalkeeper violates these rules, the penalty kick ends immediately and a goal is awarded to the attacking team.

All rules such as Ball Holding and Pushing are applied during the penalty kick. A goalkeeper will not be penalized for inactivity during a penalty kick, provided it is actively attempting to defend.

§ 14.2.2 Sudden Death Shoot-Out If teams remain tied after the initial three penalty kicks each, sudden death begins. Teams take one additional penalty kick each, and the game decision is made as follows:

Each kick is ranked in one of the following categories (from best to worst):

1. Goal scored
2. Kick at the goal that is blocked by the goalkeeper
3. Kick that hits a goalpost
4. Kick that crosses the goal line next to the goalposts
5. Any other kick
6. No kick

If both teams' kicks rank in different categories, the team with the higher ranked kick wins. Otherwise, the sudden death is repeated. If after 3 sudden death penalty kicks there is still no winner, the referee will toss a coin to decide the game.

§ 15 Kick-in / Throw-in

A kick-in (or throw-in) is an **indirect free kick** (see § 13). A kick-in is awarded to the opponents of the player who last touched the ball when the whole of the ball leaves the field of play by crossing the touchline, either on the ground or in the air. Balls are deemed to be out based on the team that last touched the ball, irrespective of who actually kicked the ball.

§ 15.1 Ball Placement

The ball is placed on the touchline at the point where it left the field.

§ 15.2 Throw-in Option

Robots may also perform a throw-in with their hands instead of kicking. When performing a throw-in with hands, the robot must:

- Face the field of play.
- Have part of each foot either on the touchline or on the ground outside the touchline.
- Hold the ball with at least one hand.
- Deliver the ball from behind and over its head.
- Release the ball within 10 seconds.

If a robot attempts to perform a throw-in with hands and fails to respect these rules, a free kick is awarded to the opposing team.

§ 15.3 Examples

Example 1: A blue robot at midfield kicks the ball over the left touchline $2\,\mathrm{m}$ into the red half of the field. The referee calls "Kick-in red" and the ball is replaced on the left touchline where it went out.

Example 2: A blue robot kicks the ball but the ball touches a red robot at midfield before leaving the field near the halfway line. The ball is regarded as out by red; the referee calls "Kick-in blue" and the ball is replaced on the touchline where it went out.

§ 15.4 Procedural Requirements

All opponents must stand at least the center circle radius (see Table 2) from the point at which the kick-in is taken. The ball is in play when it enters the field of play. After delivering the ball, the kicking robot must not touch the ball again until it has touched another player.

§ 15.5 Offenses and Sanctions

If, after the ball is in play, the kicker touches the ball again before it has touched another player, an indirect free kick is awarded to the opposing team, to be taken from the place where the infringement occurred. This rule only applies if the kicking team has three or more robots on the field at the time of the second touch.

If an opponent unfairly distracts or impedes the kicker, they are cautioned for unsporting behaviour.

For any other infringement of this rule, the kick-in is taken by a player of the opposing team.

§ 16 Goal Kick

A goal kick is a **direct free kick** (see § 13) awarded when the whole of the ball passes over the goal line, either on the ground or in the air, having last touched a player of the attacking team, and a goal is not scored.

§ 16.1 Ball Placement

The ball is placed on the corner of the goal area on the same side of the field that the ball went out. That is, the corner inside the field where the goal area line meets the goal line (not the T-junction on the goal line itself).

§ 16.2 Avoidance Region

For a goal kick, the avoidance region is the entire penalty area. All robots of the opposing team must remain outside the penalty area until the ball is in play (see § 13).

§ 16.3 Example

Example: A blue robot kicks the ball out the end of the field to the right of the goal the red team is defending. The referee calls “Goal Kick red” and the ball is placed on the right corner of the goal area.

§ 16.4 Offenses and Sanctions

If, after the ball is in play, the kicker touches the ball again before it has touched another player, an indirect free kick is awarded to the opposing team, to be taken from the place where the infringement occurred. This rule only applies if the kicking team has three or more robots on the field at the time of the second touch.

In the event of any other infringement, the kick is retaken.

§ 17 Corner Kick

A corner kick is a **direct free kick** (see § 13) awarded when the whole of the ball passes over the goal line, either on the ground or in the air, having last touched a player of the defending team, and a goal is not scored.

§ 17.1 Ball Placement

The ball is placed on the corner of the field on the same side that the ball went out.

§ 17.2 Example

Example: The red goalkeeper kicks the ball out the end of the field to the right of the goal. The referee calls “Corner Kick blue”, the ball is placed on the corner to the right of the goal, and a free kick is started.

§ 17.3 Offenses and Sanctions

If, after the ball is in play, the kicker touches the ball again before it has touched another player, an indirect free kick is awarded to the opposing team, to be taken from the place where the infringement occurred. This rule only applies if the kicking team has three or more robots on the field at the time of the second touch.

In the event of any other infringement, the kick is retaken.

IV The Official RoboCup Competition Rules

This section contains rules that are not directly relevant for games and that may not apply at local opens. However, these rules will be upheld at the yearly international RoboCup competition.

1 Referee Duty

Each team has to referee a number of games. A schedule will be released specifying the games for which each team is required to provide referees. The teams assigned to referee a game are allowed to decide among themselves which roles each team will fulfill. It is advised that the head referee and the GameController operator come from the same team.

A team may swap their scheduled refereeing duties with another team, but the team listed on the referee schedule will be held accountable if referees fail to appear for a game they are scheduled to referee.

Referees should report to the appropriate field at least ten minutes before the game is scheduled to start. If a team fails to provide the due referees for a game in which they are scheduled to provide referees, it will be noted by the Organizing Committee and recorded as a **qualification penalty**.

The requirement to referee may be an extreme hardship for extremely small teams. If a team believes providing referees for games will be an extreme hardship, they must send an email explaining their situation to the Organizing Committee and Technical Committee at least two weeks before the first set up day of the competition. The Organizing and Technical Committees will then consider the request and attempt to find an acceptable solution.

Referees must have good knowledge of the rules as applied in the tournament, and the operator of the GameController must be experienced in using that software. Referees and the GameController should be selected among the more senior members of a team, and preferably have prior experience with games in RoboCup.

In each game, each of the teams playing shall be able to veto one and only one eligible referee with no reason required. The veto must be delivered before the start of the game (the first `ready` state) or during a timeout.

The selection of referees, including the application of the above rules, must always be made under the constraint that no member of a team may be allowed to referee a game played by their own team under any circumstances.

2 Source Code Releases

All teams that have participated in RoboCup are strongly encouraged to subsequently release the code from that year's codebase. If they do, the code must be licensed such that other RoboCup participants can use it, although the license may place conditions on its use.

The preferred type of release is the full source code of the software that was running in the team's last game at RoboCup. In case this is not possible (e. g. due to legal reasons), it is required that at least the source code related to the novel contributions (as given during the qualification process) is published. Participation in technical challenges or research challenges may come with additional requirements on the amount of components to be released.

The source code must be published and its availability announced on the league mailing list (`PlaceholderforLeagueEmail`) by 2026-10-15. Failing to publish source code by the deadline will result in a qualification penalty being recorded.

This will become a requirement in the following years.

3 Rules for Forfeiting

Teams who do not make a good faith effort to participate in a scheduled game are considered to forfeit the game.

If a team notifies the technical committee that they wish to forfeit less than two hours before their scheduled game time, simply fails to show up for their game, or decides during their game that they wish to forfeit, then the opposing team will play the match against an empty field. However, any own goals will not be scored. Hence, after an opponent forfeits, the team playing against an empty field cannot do worse than they were doing at the time the opponent decided to forfeit. Teams may choose to forfeit at any stoppage of play. However, once a forfeit is announced, they may not reverse this decision.

If a team notifies the technical committee that they wish to forfeit at least two hours before their schedule game time, the following procedure will be followed.

- If a team chooses to forfeit a match in the round robin games the other team plays the match against an empty field. However, any own goals will not be scored.
- If a team chooses to forfeit in a knock-out game it gets replaced by the next best qualified team, i. e. the team it kicked out or left behind in the round robins.

Note that there are a few unlikely cases that are not covered by these rules. If a situation is not covered by these rules, the technical committee and the organizing committee will work together to make a decision.

Any forfeit will result in a qualification penalty being recorded but the circumstances of the forfeit will affect the severity of the offense and the impact on future qualification.

4 Disqualification During Competition

A team may be disqualified during the RoboCup competition for:

- Receiving a red card assigned to their team as humans (see § 12)
- A serious violation of the terms of a team's qualification
- Gaining a Qualification Penalty during the course of the competition
- A serious breach of ethics, or serious behavior unbecoming of participants of RoboCup.

Example. A team promises to use their novel contribution in RoboCup games, but fails to do so. Alternatively, a team deliberately misleads the technical committee about the novelty of their work and/or their contribution to the league, such that they are deemed to have copied another team.

A team can *only* be disqualified by a decision of the *Board of Trustees of the RoboCup Federation*. The RoboCup Soccer League executive must petition the board in writing at their soonest possible availability. The executive must simultaneously inform the relevant team of the petition in writing.

A disqualified team automatically forfeits all games (see Section 3). For practicality, the disqualification should not apply *retroactively*. However, by majority vote of the team leaders, provisions for retroactive disqualification may be made in the fairness of the affected teams.

5 Awards

5.1 Best Referee Voting

Teams will be asked to vote for the team who has provided the best referrees throughout the competition. Specifics on the voting process and awards will be detailed by the Organizing Committee during a referree meeting before the start of the competition.

5.2 Best Humanoid Award

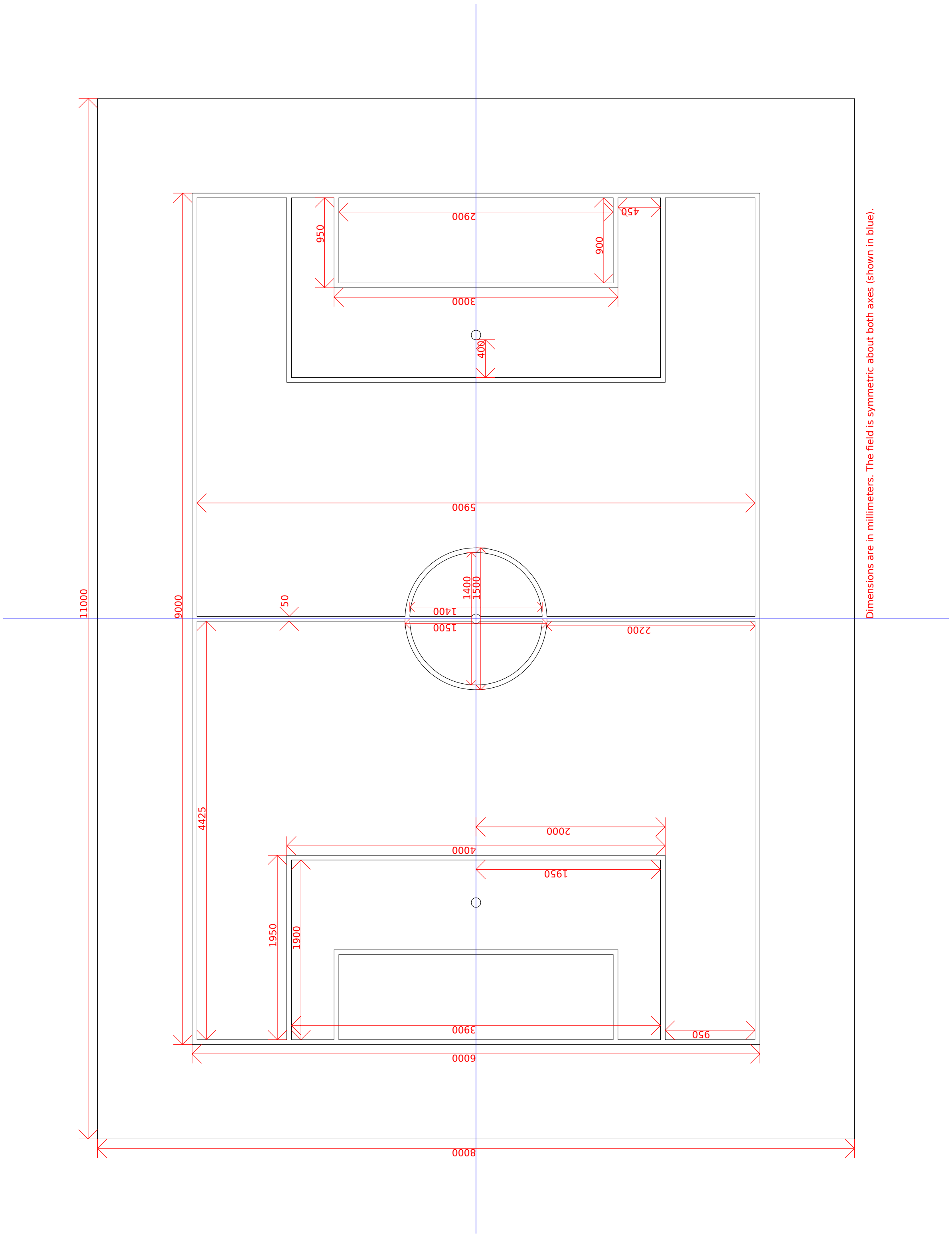
TBA

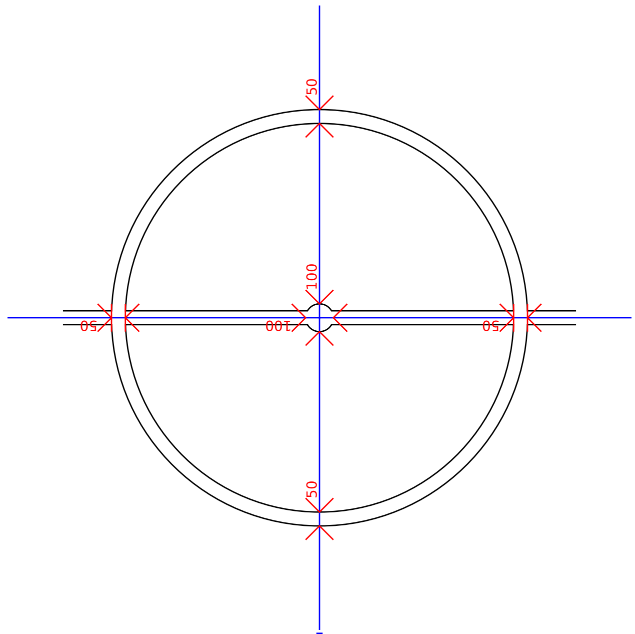
V Law Changes

These rules were designed for RoboCup 2026 and the newly founded Humanoid Soccer League. They are based on the rule-books of the former Standard Platform League, Humanoid League KidSize and Humanoid League AdultSize. This section is reserved for future changes.

A Field Technical Drawings

1 S-Field

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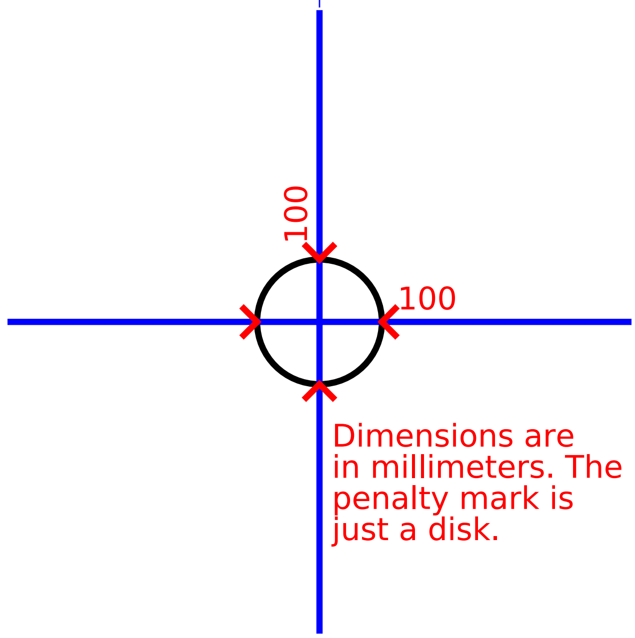
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Dimensions are
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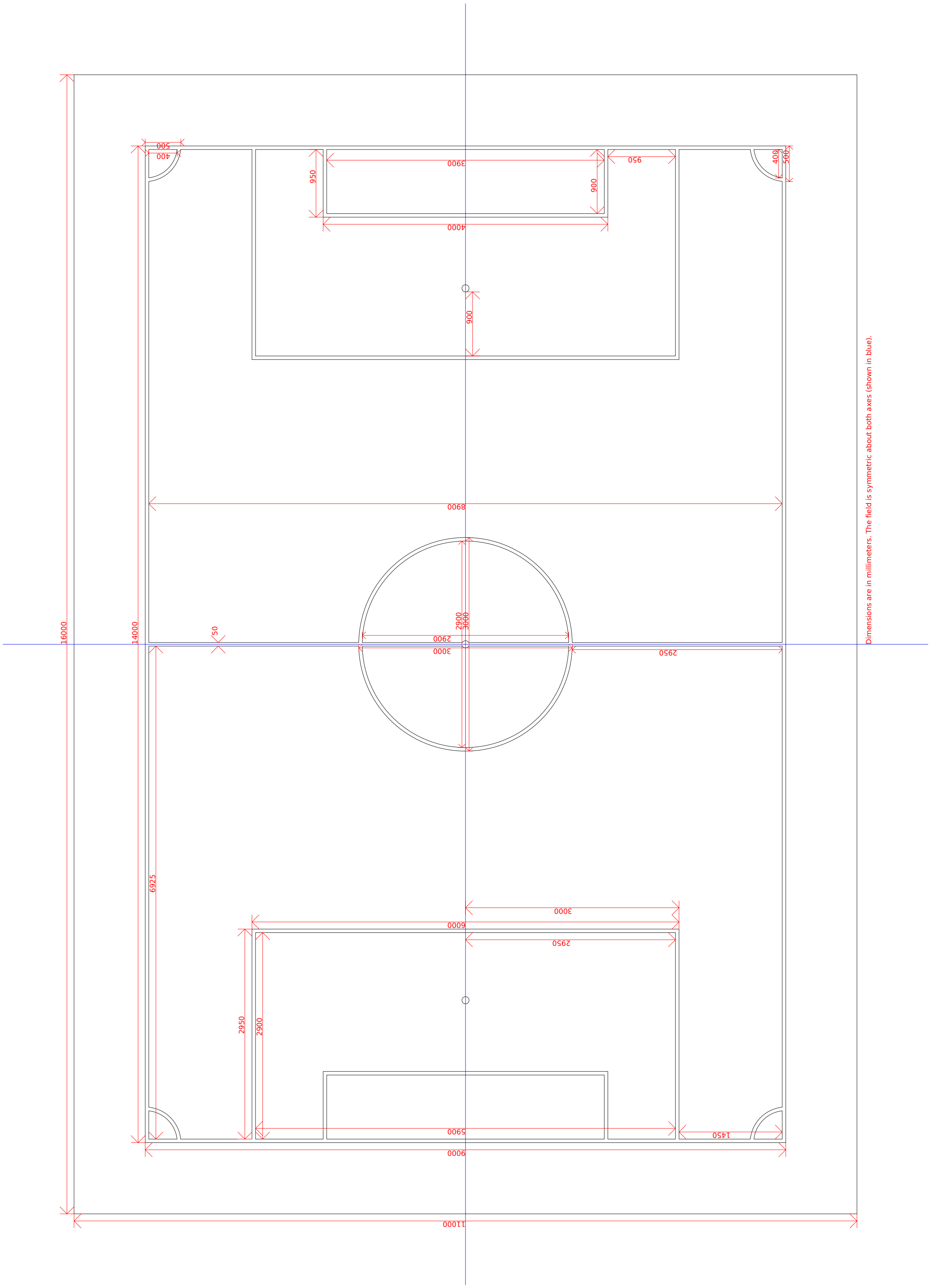
penalty mark is
just a disk.

2 M-Field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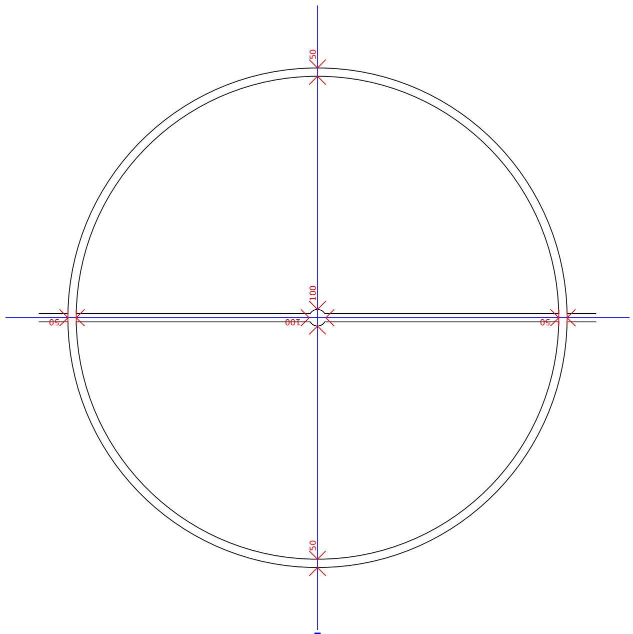
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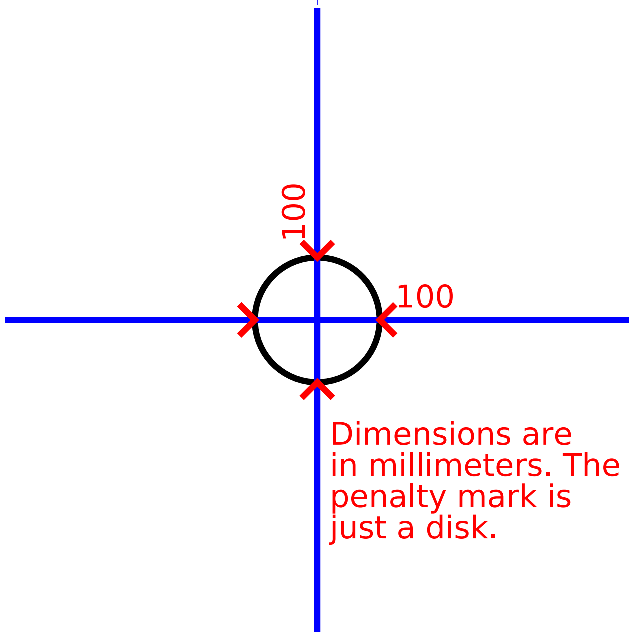
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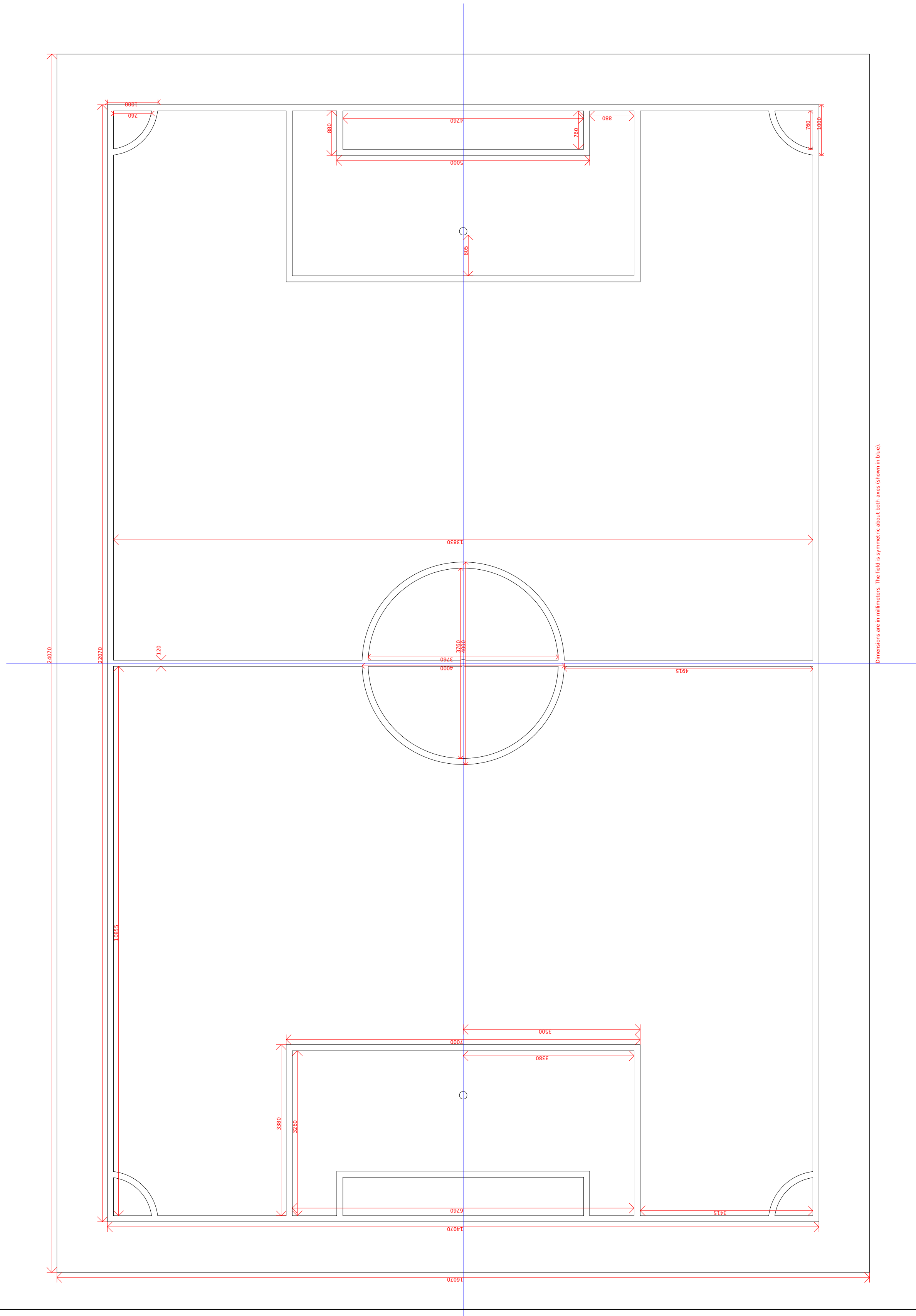
Dimensions are in millimeters. The penalty mark is just a disk.

3 L-Field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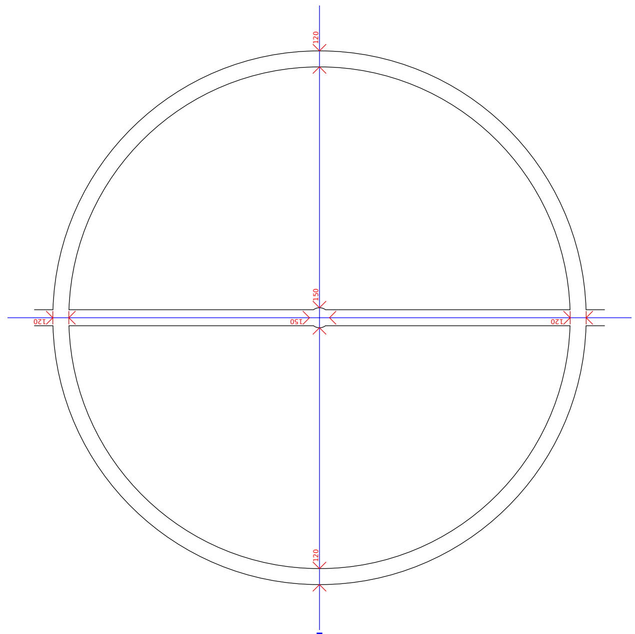

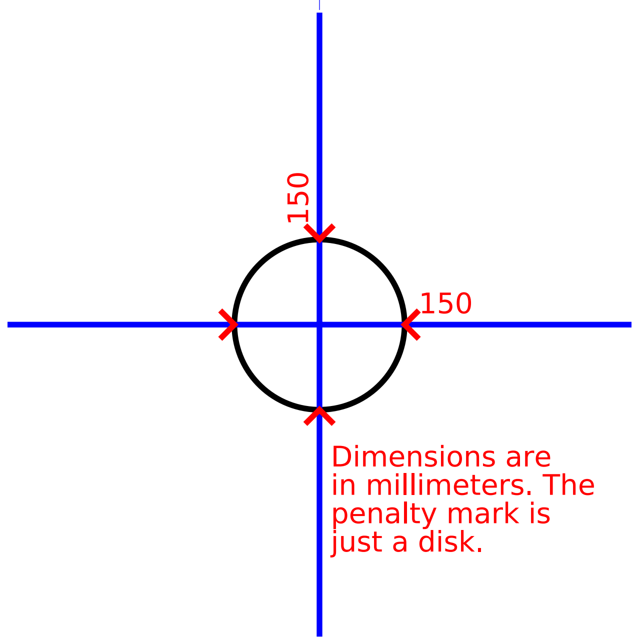
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in millimeters. The
penalty mark is
just a disk.